JÖNKÖPING UNIVERSITY
*School of Engineering*

# One Platform, Many Chairs

Designing an ergonomic office chair based on a shared product platform.

**PAPER WITHIN** *Product Design*

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The authors take full responsibility for opinions, conclusions and findings presented.

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# Abstract

Office chairs are used in increasingly flexible and varied work environments, where users may have different ergonomic needs, work habits and aesthetic preferences. At the same time, manufacturers need to create product collections that allow variation without making the product structure too complex. This thesis was carried out in collaboration with Lanab Group AB and explores how a shared product platform can be used as a foundation for designing an office chair with integrated ergonomic functions and visual variation.

The aim of the project was to identify which ergonomic functions are essential in an office chair platform and to develop a design concept based on a shared technical base. The study was conducted through a combination of literature review, user questionnaire, user observations, interviews and assembly observation at Lanab. These methods were used to understand ergonomic requirements, user interaction with chair functions, and how different chair components are connected in the assembly structure.

The results showed that functions such as seat height, seat depth, backrest tilt, rocking function, tilt tension and seat angle were strongly connected to the chair mechanism and could therefore be understood as core platform functions. The backrest was also identified as an important part of the platform structure, although adjustable backrest height was not directly integrated into the core mechanism. The assembly observation further showed that the mechanism acts as a central connection point between the seat, backrest, armrests, gas lift and base, making it a suitable foundation for a shared office chair platform. Based on these findings, a final concept was developed where the mechanism and its main connection points form the shared product platform. The design was then explored through different chair variations, including a mesh backrest version and a foam backrest version, to show how the same platform can support variation in expression, comfort and product identity. The final concept demonstrates how ergonomic functionality and design variation can be combined within a shared platform structure.

The thesis concludes that a shared product platform can support both ergonomic consistency and product variation when the central functions and connection points are clearly defined. This approach can help create an office chair collection that is flexible, visually adaptable and easier to develop around a common technical base.



Keywords: office chair, product platform, ergonomics, modular design, product variation, industrial design.

# Sammanfattning

Kontorsstolar används idag i arbetsmiljöer som blir allt mer flexibla och varierade. Det gör att stolar behöver kunna möta olika användares ergonomiska behov, arbetssätt och estetiska preferenser. Samtidigt behöver tillverkare kunna skapa produktkollektioner med variation, utan att konstruktionen blir för komplex. Detta examensarbete har genomförts i samarbete med Lanab Group AB och undersöker hur en gemensam produktplattform kan användas som grund för att designa en kontorsstol med ergonomiska funktioner och visuell variation.

Syftet med projektet var att identifiera vilka ergonomiska funktioner som är viktiga i en kontorsstolsplattform och att utveckla ett designkoncept baserat på en gemensam teknisk grund. Arbetet genomfördes genom en litteraturstudie, en användarenkät, användarobservationer, intervjuer och en monteringsobservation hos Lanab. Metoderna användes för att förstå vilka ergonomiska behov som finns, hur användare faktiskt interagerar med stolens funktioner och hur olika delar av stolen är kopplade till varandra i monteringen.

Resultatet visade att funktioner som sitthöjd, sittdjup, ryggstödslutning, gungfunktion, viktjustering och sitsvinkel hade en tydlig koppling till stolens mekanism. Dessa funktioner kunde därför ses som centrala funktioner i plattformen. Ryggstödet identifierades också som en viktig del av plattformsstrukturen, även om justerbar ryggstödshöjd inte var direkt integrerad i själva huvudmekanismen. Monteringsobservationen visade dessutom att mekanismen fungerar som en central kopplingspunkt mellan sits, ryggstöd, armstöd, gaspelare och fotkryss. Detta gjorde mekanismen till en lämplig grund för en gemensam produktplattform. Utifrån dessa insikter utvecklades ett slutkoncept där mekanismen och dess huvudsakliga kopplingspunkter bildar den gemensamma produktplattformen. Designen utforskades sedan genom olika stolvariationer, bland annat en version med nätrygg och en version med stoppad rygg. Detta gjordes för att visa hur samma plattform kan skapa variation i uttryck, komfort och produktidentitet, samtidigt som de viktigaste ergonomiska funktionerna behålls.

Examensarbetet visar att en gemensam produktplattform kan stödja både ergonomisk funktionalitet och produktvariation när centrala funktioner och kopplingspunkter definieras tydligt. På så sätt kan plattformen bidra till en kontorsstolskollektion som är flexibel, visuellt anpassningsbar och enklare att vidareutveckla utifrån en gemensam teknisk grund.



Nyckelord: kontorsstol, produktplattform, ergonomi, modulär design, produktvariation, industriell design.

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# Declaration of AI Use

Artificial intelligence tools have been used during the writing process of this thesis. The tools were mainly used to support spelling correction, improve sentence structure and refine certain formulations.The content, analysis, design work and final decisions presented in this thesis are my own. AI tools were used only as support in the writing process and not as a replacement for my own work, interpretations or conclusions.

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# 1. Introduction

This chapter introduces the project and provides an overview of its context, purpose and direction.

## 1.1. Background

In contemporary work environments, a growing number of people spend a significant portion of their day sitting at their workplaces. As work becomes increasingly desk-based, the office chair plays a crucial role in supporting users' health, comfort, and long-term well-being [1]. To respond to these demands the office furniture market offers a wide range of office chairs in different forms, colors and configurations. These variations aim to address both ergonomic requirements and aesthetic preferences across diverse office environments. From a user perspective, this diversity provides choice and adaptability [2]. From a manufacturing perspective, however, offering many chair models and variations introduces additional challenges. Product variation often requires differences in components, functions, and structures, which increases product complexity and affects manufacturing efficiency [3]. As a result, the development of ergonomic office chairs involves balancing user needs, market expectations, and production-related constraints.

To manage increasing product variety while maintaining efficiency, product platform strategies have been widely adopted across several industries. A product platform can be defined as a shared set of components and processes that enables companies to efficiently develop differentiated products through a flexible and responsive development process [4] [5].Platform-based product development has proven successful in industries such as automotive, electronics and domestic appliances, where it supports standardisation, reduced lead time and cost efficiency [5] [6]. By combining platform strategies with concepts such as mass customisation, modularisation, and product configuration, companies can offer product variety while maintaining internal efficiency [5] [2].

While product platform strategies offer clear advantages in terms of standardisation, cost reduction and efficient product family development, their application within the design of ergonomic office chairs introduces additional challenges. Office chairs often include multiple adjustable mechanisms intended to support different users and working positions. However, research shows that the presence of ergonomic features does not automatically lead to improved comfort or effective use. Without guidance or instruction, users may evaluate chairs based on first impressions rather than correctly adjusting and experiencing the ergonomic functions. In some cases, ergonomically inferior chairs have even been perceived as more comfortable than superior ones during unguided exploration[7]. This suggests that simply increasing the number of adjustable features to achieve product variation does not guarantee better ergonomic outcomes in practice. Instead, it raises questions about which functions are truly essential and how ergonomic potential can be realised in a meaningful and usable way.

## 1.2. Problem Description

The design and development of ergonomic office chairs involve a complex set of requirements that must be addressed at the same time. Office chairs are expected to support users health and comfort during long periods of seated work which has led to the integration of numerous adjustable functions intended to accommodate

different body types and working behaviors [1]. However, as the number of functions increases, so does the structural and mechanical complexity of the product. Additional mechanisms, components, and structural variations can make chairs more difficult and costly to manufacture, particularly when companies aim to produce multiple models or product lines [3]. At the same time, companies operate in a competitive market where visual differentiation, new collections and functional updates are necessary to meet customer expectations [2].

Developing separate mechanisms or unique structures for each chair model reduces opportunities for component reuse and increases production complexity. This limits scalability and goes against the principles of platform-based product development, where variation is created through shared components and structures [4][5].

This challenge can be observed within the Swedish office seating company Lanab Group AB, which offers a broad product portfolio consisting of numerous seating models. While this allows the company to address different user needs and market segments, many of the chairs are based on different mechanisms and structural configurations. As a result, the product portfolio includes many unique components and mechanisms, which increases production complexity and limits opportunities for efficient large-scale manufacturing. To address these challenges, there is a need to explore how office chairs can be structured around a more standardized product architecture. A shared product platform could potentially enable multiple chair variants to be developed from a common structural base, allowing product variation while reducing structural complexity. Investigating how such a platform can support ergonomic functionality while enabling variation within a product collection forms the basis of this thesis.

## 1.3. Objective and Aim

The objective of this thesis is to develop a modular office chair concept based on a shared product platform, with focus on how essential ergonomic functions can be integrated, positioned and expressed within a coherent chair architecture. The study aims to identify which adjustable functions are necessary across all chair variants and explore how the placement and design of these adjustment mechanisms can support both ergonomic performance and platform flexibility. Through a design-driven process informed by ergonomic research, user needs, stakeholder input and product platform theory, the project investigates how a shared structural base can support variation in form and configuration while maintaining ergonomic functionality and a unified product structure. The outcome is a concept proposal that demonstrates how platform thinking can be applied within the design of ergonomic office chairs.

## 1.4. Research Questions

To fulfil the objective of this study, the following research questions are investigated:

1. *Which ergonomic functions are essential when designing a shared product platform for office chairs?*

**2.** *How can an office chair be designed based on a shared product platform to support product variation?*

## 1.5. Scope and Delimitation

This study focuses on the design of a product platform for ergonomic office chairs within the field of industrial design. The scope is limited to identifying essential ergonomic functions and exploring how these can be structured within a shared product platform that enables multiple chair variants. The work primarily addresses product architecture, modular structuring, and conceptual development, rather than detailed mechanical engineering or full-scale industrial implementation.

The study does not include the development of specific manufacturing processes, production cost analyses, or efficiency calculations. While general manufacturing considerations inform the platform structure, technical production simulations and financial evaluations are outside the scope of this thesis.

Furthermore, the outcome of this project is a concept-based design proposal that demonstrates how a modular office chair can be developed from one coherent product platform. Ergonomic considerations are grounded in existing research and established ergonomic principles, and no medical or biomechanical testing is conducted as part of this work. The mechanical solutions developed in this study are limited by existing measurements, component dimensions and technical requirements provided by the company. This means that the creative freedom in the mechanical development was partly restricted, as the platform needed to relate to the company's existing chair structure and connection points. Therefore, the mechanical solutions should be understood as concept-level proposals rather than fully engineered technical solutions.

## 1.6. Contribution to the field

This thesis contributes to the field of industrial design by showing how product platform thinking can be applied as a design strategy in the development of ergonomic office chairs. The study adds to the understanding of how a shared product platform can be defined not only through technical components, but also through ergonomic functions and design-related requirements.

The main contribution is the connection between ergonomic function, platform structure and product variation within office chair design. By identifying which ergonomic functions should be part of the shared platform, the study shows how ergonomic consistency can be maintained across different chair variants. At the same time, it demonstrates how variation in form, material expression and backrest design can be created without changing the core platform.

The study also contributes by translating findings from literature, user input, observations, stakeholder needs and trend analysis into design requirements for a platform-based chair concept. In this way, the thesis provides an example of how industrial design can support platform development by balancing technical structure, ergonomic performance and visual variation in one design process.

## 1.7. Disposition

This thesis report is divided into seven main chapters:

1. *Introduction* – This chapter presents the background, problem description, objective and aim of the project. It also includes the research questions, scope and delimitation, contribution to the field and the disposition of the report.
2. *Theoretical framework* – This chapter presents the theoretical areas that support the project, including design for emotion, product platform theory, ergonomics in work environments and seating ergonomics.
3. *Methodology* – This chapter describes the methods used in the project, structured according to the Double Diamond design framework. The methods are divided into the phases discover, define, develop and deliver.
4. Implementation and result for methods – This chapter presents how the methods were carried out and the results from each phase. It includes findings from the literature review, benchmarking, questionnaire, observations, interview, trend spotting, personas, design direction, concept selection, brainstorming and physical model.
5. *Final concept result* – This chapter presents the final design concept, including the core platform, the backrest attachment and the overall office chair design.
6. *Discussion* – This chapter discusses the results in relation to the research questions. It reflects on the essential ergonomic functions, the design language, user interaction, platform structure and product variation.
7. *Conclusion* – This chapter presents the main conclusions of the thesis, followed by future scope and personal reflection.

The report ends with references and appendix, where supporting material and additional information are presented.

# 2. Theoretical framework

This chapter presents the theoretical foundation used to support and frame this study.

## 2.1. Product Platform Theory

A product platform is described as a shared base that makes it possible to develop several product variants from the same overall structure. Instead of designing every product from the beginning, a company can reuse common parts, processes, knowledge and solutions across a product family. This helps companies create variation while keeping control over complexity, cost and development time. Product platforms are therefore often connected to modularity and product families, because they make it possible to combine standardized parts with adaptable solutions. In this way, a product platform supports both efficiency and flexibility in product development [5] [6].

In design, a product platform is not only about reusing physical components, but also about creating a clear structure for how different products can be developed. The platform can define which functions, interfaces and technical solutions should be shared between products, and which parts can change to create different variants. For an office chair, this means that important ergonomic functions and structural elements can be integrated into one common base, while aspects such as form, materials, proportions and visual expression can vary between different chair models. A product platform can therefore be seen as a design strategy that helps create a coherent product family while still allowing variation and individual design solutions [5] [6].

## 2.2. Ergonomics in Work Environments

Ergonomics is about how people interact with their work environment, tasks, tools, and systems. In a workplace, the aim of ergonomics is to make the work fit the person, instead of forcing the person to adapt to poor working conditions. This means that the workplace, equipment, furniture, and work routines should be designed to support people's comfort, safety, health, and performance. When ergonomics is not considered, employees may experience discomfort, fatigue, stress, or work-related injuries. A well-designed work environment can instead help people work in a safer, more comfortable, and more effective way [8].

Work ergonomics can be understood through three main perspectives: physical ergonomics, organisational ergonomics, and cognitive ergonomics. *Physical* concerns the relationship between the human body and the physical conditions of work. It includes aspects such as posture, movement, body dimensions, physical load, workplace layout and the design of furniture, tools, and equipment. The purpose of physical ergonomics is to reduce physical strain and support the body during work tasks.

*Cognitive* ergonomics is about how people understand information, make decisions, react, learn, and interact with other people or digital systems in the workplace. In simple terms, it looks at how work affects the mind, not only the body. This can include workload, stress, training, decision-making, attention, perception, and the relationship between humans and systems. If a task is unclear, if the user does not have enough training, or if too much information needs to be handled at once, the work can become mentally demanding. Cognitive

ergonomics therefore aims to reduce unnecessary mental strain and make work easier to understand, manage, and perform[8].

*Organisational* ergonomics focuses on how work is planned, structured, and managed. This includes communication, teamwork, work routines, job design, organisational culture, policies, and how responsibilities are divided. This perspective shows that ergonomics is not only affected by physical products, but also by the broader context in which work takes place. A work environment can therefore support ergonomics when the organisation, routines, and systems make it easier for people to perform their tasks in a safe and effective way.

## 2.3. Seating Ergonomic

Sitting ergonomics focuses on the relationship between the user, the chair, and the working environment, with the aim of reducing physical strain and supporting comfortable and efficient work. Since office workers often spend a large portion of the day seated, prolonged static sitting is associated with musculoskeletal discomfort and fatigue. Proper sitting ergonomics therefore require that the chair supports the user's body dimensions and allows posture variation during sitting [9] [10].

A neutral sitting posture is typically defined as one that maintains the body's natural alignment. In this posture, the spine retains its natural S-shaped curvature, with an inward curve in the lower back (lumbar region) and support in the mid-back (thoracic region). This is commonly associated with a slight anterior pelvic tilt (pelvis tilts forward), which helps maintain the lumbar curve [9] [11]. The head, shoulders, and hips should be approximately aligned vertically, while the shoulders remain relaxed. The feet should rest flat on the floor, with the thighs approximately parallel to the ground and the knees positioned at or slightly below hip level. In a working posture, the elbows are typically positioned around 90 degrees, allowing the forearms to rest comfortably near the work surface[9].

However, maintaining a static posture for prolonged periods may lead to muscle fatigue. Sitting ergonomics therefore emphasizes posture variation and dynamic sitting. Small movements of the spine and pelvis (the body segment connecting the upper and lower body) may help redistribute pressure, reduce static muscle loading, and support circulation. Adjustable chair functions that allow movement and changes in posture can therefore contribute to improved sitting ergonomics during prolonged seated work [11] [12] [13].

### 2.3.1. Ergonomic support controls in Office Chair

#### 2.3.1.1. Lower body support

Lower body support refers to how the chair supports the pelvis, hips, thighs, knees, lower legs and feet during sitting. This support is important because the lower body affects pelvic position, spinal alignment and pressure distribution across the seat.[14] [13] [15]

##### Seat height

One important control to achieve a neutral lower body position is the Seat height adjustment in office chairs. An adjustable seat height allows users to place their feet flat on the floor while maintaining appropriate knee and [13] [15]. If the seat is too high, the thighs can be pushed upward, and the feet may not rest properly on the floor. This increases pressure under the thighs and reduces stability in the sitting posture. Because the lower body is not well supported, the user may compensate by leaning forward or adjusting the upper body position. This can affect the natural curvature of the spine and lead to a less supported posture. In addition, when the seat height does not match the desk height, the user may raise the shoulders or bend the neck to reach the work surface, which can increase strain in the upper body. If the seat is too low, the knees are positioned higher than the hips, which can cause the pelvis to tilt backwards. This reduces the natural curve of the lower back and often results in a more rounded sitting posture. Over time, this posture may reduce support and increase discomfort during prolonged sitting [12] [11] [16] [17] [18] [19] [20].

## Seat depth

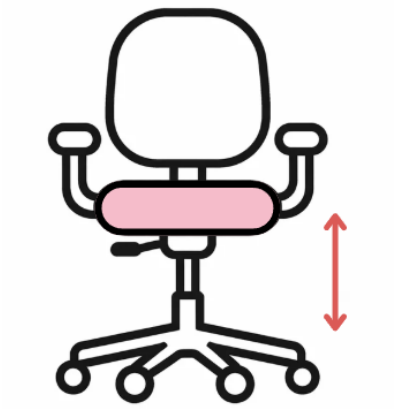


***Figure 1:*** *Illustration of seat height adjustment.*

Seat depth is an adjustable control in office chair design that allows the seat to be adapted to the user's buttock–popliteal length. Proper adjustment of seat depth is important to support the lower body and enable contact with the backrest. An ergonomic mismatch may occur when the seat depth is shorter than approximately 80% or longer than 95% of this body measurement. Seat depth mainly affects the lumbar region, the thighs, and the popliteal area behind the knees, where sensitive blood vessels and nerves are located [12]. When properly adjusted, seat depth allows the user to sit fully against the backrest, supporting the lumbar region and helping to maintain the natural S-shaped curvature of the spine. In this position, the pelvis may remain slightly tilted forward, the thighs are supported along a larger contact area, and the feet can rest flat on the floor. This alignment may contribute to improved pressure distribution and stability in the lower body. If the seat is too deep (too far from the backrest), the user may be prevented from using the backrest, resulting in reduced lumbar support. This may affect spinal alignment and increase pressure under the thighs, particularly behind the knees, which may restrict circulation. If the seat is too shallow (too close to the backrest), thigh support is reduced and the contact area becomes smaller, which may lead to decreased stability and uneven pressure distribution during sitting [15] [12] [11] [17] [18] [19] [20] .

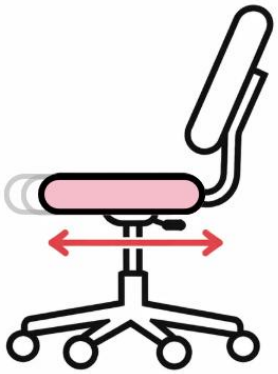


***Figure 2:*** *Illustration of seat depth adjustment.*

### Seat angle

Seat angle is an adjustable control that influences pelvic position and overall sitting posture. By tilting the seat, the orientation of the pelvis changes, which affects spinal alignment. A slightly forward-tilted seat may support an anterior pelvic tilt and help maintain the natural S-shaped curvature of the spine. In contrast, a flat or backward-tilted seat may increase the tendency to slouch, as the pelvis rotates backwards and the spine becomes more flexed [13] [15] [11] [19]. A horizontal seat (0°) is often used as a mechanical neutral seat angle position, but this does not necessarily correspond to a biomechanically neutral posture. A flat seat may promote posterior pelvic rotation (pelvis tilts backward), while a slightly forward-tilted seat may better support anterior pelvic tilt (pelvis tilts forward) and a more upright sitting posture [12] [16].

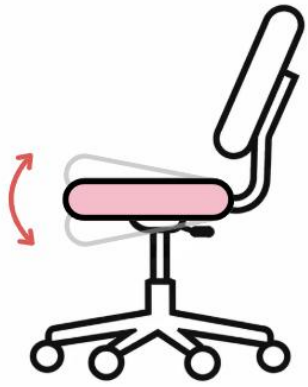


***Figure 3:*** *Illustration of seat angle adjustment.*

## 2.3.1.2. Upper body support

Upper body support refers to how the chair supports the lower back, upper back, shoulders, neck and arms during sitting. This support is important because the upper body affects spinal alignment, muscular load and the ability to maintain a supported working posture [11] [12] [13].

### Backrest height

Backrest height is an adjustable control that ensures the backrest aligns with the user's spinal curvature. Proper adjustment allows the backrest to support the lumbar and thoracic regions (chest area located between the neck and abdomen), helping maintain the natural S-shaped curvature of the spine. Adequate support may reduce

muscular effort and decrease loading on the intervertebral discs during prolonged sitting [13] [20] [21] [22] [23] [24] [25].

If the backrest is too low (insufficient height), support for the lumbar and thoracic regions is reduced. This may lead to pelvis tilting backward, flattening of the lumbar curve and a more flexed spinal posture. Such alignment may increase compressive forces on the intervertebral discs and require greater muscular activity to stabilize the trunk. If the backrest is too high (extends into the shoulder blade area), movement of the upper body may be restricted. This can interfere with shoulder motion and may contribute to increased strain in the neck and upper back. Adjustable backrest height therefore allows the support areas of the chair to align with the user's spine without limiting upper body movement [11] [12] [13] [17] [19] [21] [22] .



***Figure 4:*** *Illustration of backrest height adjustment.*

## Lumbar Support

Lumbar support is an adjustable control designed to support the lower spine, prevent the pelvis tilting backward and to maintain the spines S-shape during prolonged sitting. The main purpose of lumbar support is to preserve the lumbar lordosis (inward curve of the lower back), which tends to flatten when sitting without a lumbar support. Proper lumbar support may reduce compressive forces on the intervertebral discs in the lower spine and decrease the muscular effort required to keep the upper body upright. By supporting the lower back, part of the upper body weight can be supported by the backrest instead of being carried only by the spine and surrounding muscles. This may contribute to a more even pressure distribution between the seat and the backrest [12] [11] [25].

The effectiveness of lumbar support depends on how well it matches the user's lower back. Fixed lumbar supports may not align with the curvature of the lower spine, which can leave the lumbar region unsupported. Adjustable lumbar support, typically in height and sometimes depth, allows the support to be positioned according to the user's spinal curvature. This may provide more targeted support of the lower back and help maintain spinal alignment during prolonged sitting [13] [10].



***Figure 5:*** *Illustration of lumbar support.*

## Armrest

Armrests are adjustable controls designed to support the weight of the arms and reduce loading on the shoulders, neck, and upper back. By supporting the forearms, the muscular effort required to hold the arms in a working position may be reduced, which can contribute to a more relaxed upper body posture during seated tasks [10]. Armrests may also contribute to maintaining an upright sitting posture by providing support for the upper body. Since the upper body naturally tends to collapse forward during prolonged sitting, armrests can be used to stabilize the posture by allowing the user to lean the forearms against the armrests for support [16].

Armrest height influences shoulder position [10]. If the armrests are too high (push shoulders upward), the shoulders may become elevated, which can lead to tension in the neck and upper back. If the armrests are too low (insufficient arm support), the user may lean forward or slump to reach the support, which may affect spinal alignment and increase loading on the lower back [3] [13] [17] [18]. A neutral position is when the elbows are at approximately 90 degree flexion,  typically achieved when the armrests support the elbows with relaxed shoulders and the forearms approximately parallel to the work surface [13] [12] [17].

Additional adjustments, such as width, length, and angle, allow the armrests to be adapted to different body dimensions and tasks. If the armrests are too wide (far from the body), the arms may move away from the torso, increasing shoulder loading. If the armrests are too narrow (too close to the body), they may restrict movement or compress the hips [3]. Armrest length also influences posture. If the armrests are too long the user may be prevented from sitting close to the work surface, which can lead to forward leaning and reduced backrest

support [13]. Adjustable angle allows the armrests to follow the natural direction of the arms during different tasks, which may help maintain relaxed  shoulders and reduce strain in the upper body [3] [10].



*Figure 6: Illustration of armrest adjustability, including height, depth and angle adjustment.*

### 2.3.1.3. Dynamic seating mechanisms

Dynamic seating support refers to how the chair allows the body to move and shift position while sitting. This support is important because small movements of the spine, pelvis and upper body can reduce static loading and support posture variation during the workday.[12] [11] [10].

#### Lockable backrest tilt

Lockable backrest tilt is an adjustable control that allows the backrest to be fixed at a selected angle, providing continuous support for the thoracic and lumbar regions of the spine. By locking the backrest in a supportive position, the upper body can rest against the backrest, which may reduce muscular effort in the lower back and help maintain a neutral spinal alignment during prolonged sitting [12] [10] [17] [19].

If the backrest angle is too upright the pelvis may rotate backward which can flatten the lumbar curve and increase loading on the lower spine. This may contribute to a more slouched posture and increased muscular strain[13] [12] [19]. If the backrest is too far reclined the user may need to bend the neck forward to view the work surface, which can increase tension in the neck and shoulders. Excessive recline may also cause the user to slide forward, reducing contact with the lumbar support [3] [12] [19].

A rigid 90-degree sitting position is often insufficient during prolonged sitting. A slightly reclined backrest angle, typically around 95-110 degrees, may better support the natural spinal curvature and reduce loading on the spine. The ability to adjust and lock the backrest also enables posture variation, which allows small movements of the spine and pelvis. Such variation may help distribute pressure, reduce static muscle loading, and support more dynamic sitting over time [11] [10] [17].



***Figure 7:*** *Illustration of lockable backrest tilt.*

## Rocking motion

Rocking motion is an adjustable control that enables dynamic sitting by allowing the backrest or seat to move in response to the user's body movements. This function supports continuous posture adjustments and micro-movements of the spine and pelvis, allowing the user to shift between a more upright and a more reclined position. Such movement helps avoid prolonged static sitting, which is often associated with posterior pelvic rotation and flattening of the lumbar curve [11] [26] [27] [23].

The rocking function also contributes to pressure redistribution while sitting. As the chair tilts, part of the body load can shift from the sitting bones toward the thighs and backrest, which may reduce concentrated pressure and support a more even load distribution [12]. These small posture changes may reduce static muscle loading and help maintain the natural S-shaped curvature of the spine during prolonged sitting [12] [11]. Different rocking mechanisms support dynamic sitting in different ways .For focused tasks, rocking functions are often combined with a lockable tilt, allowing the user to switch between dynamic movement and a stable working posture [7] [13] [12] [27] [23] .

In relation to the rocking function, tilt tension control can be used to adjust the resistance of the movement according to the user's body weight and preference. This allows the chair to respond more appropriately to the user's movement while still providing a sense of stability during seated work [28].

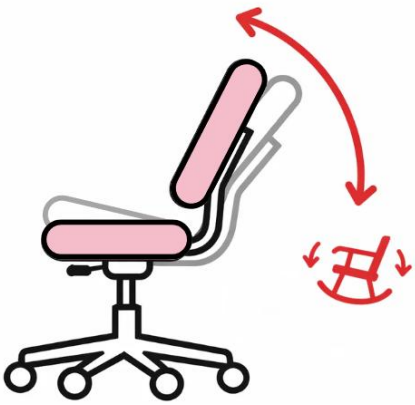


***Figure 8:*** *Illustration of the rocking function, showing the chair's ability to move back and forth.*

# 3. Methodology

This chapter describes the methodological approach used to guide the study process.

## 3.1. Double diamond design framework

The Double Diamond is a design framework developed by The Design Council that illustrates the design and innovation prosses between exploring a problem and narrowing down towards a defined solution. The framework structures design work into four phases: Discover, Define, Develop, and Deliver [29]:

The first phase, Discover, focuses on understanding the context and identifying user needs. During this stage, insights are gathered through the aim is to explore the problem area and gain a broader understanding of users, needs, and requirements [30]. The second phase, *Define*, involves analysing the collected material. Key insights are identified the design challenge is defined [31]. The third phase, *Develop*, focuses on generating and exploring possible solutions [32]. The final phase, Deliver, involves refining and presenting the solution. The outcome is a final design proposal supported by visualizations and documentation [33].

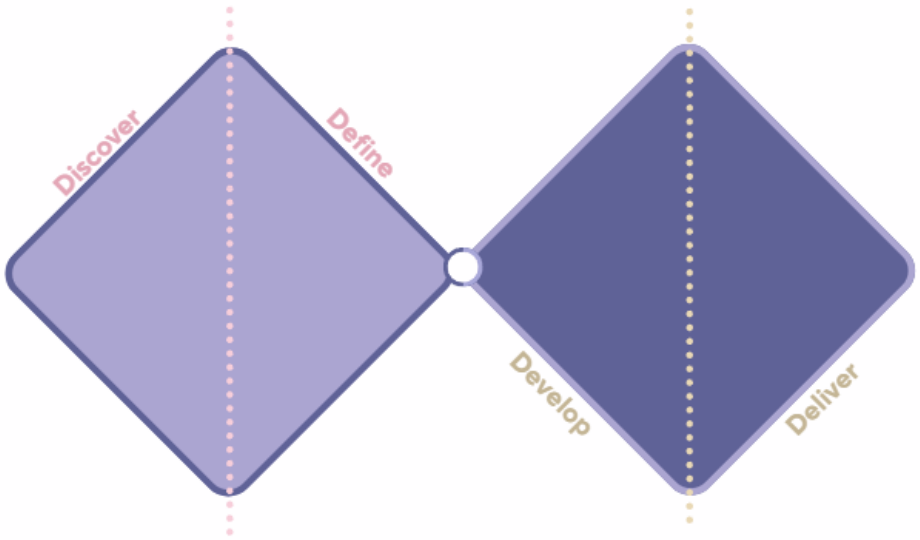



***Figure 9:*** *The Double Diamond design process, including the Discover, Define, Develop and Deliver phases.*

## 3.2. Discover

The Discover phase focuses on exploring the problem area and gaining a broader understanding of the context. During this phase, information is collected to identify user needs, challenges, requirements and possible opportunities[30].

### 3.2.1. Literature Review

A literature review is a research method used to examine existing knowledge within a specific field. In design research, literature reviews can help identify key issues, relevant theories and previous findings that provide a foundation for the design process [34] [35].

In this study, the literature review was used to evaluate ergonomic office chair functions based on existing research. The method was relevant for understanding which adjustable functions are supported in the literature and how these functions relate to ergonomic sitting. The selected literature was structured and compared to identify recurring support for specific functions. This supported Research Question 1 by contributing to the understanding of which ergonomic functions may be considered essential for an office chair.

### 3.2.2. Competitive Benchmarking analysis

Competitive benchmarking is a method used to compare existing products within a market based on predefined criteria. The method can help identify common product features, differences between products, and possible development opportunities. This gives a clearer picture of the existing market, and the advantage and disadvantage of each product [35]. Competitive benchmarking is also used to compare products against defined metrics, providing a structured basis for design decisions and product positioning [36].

In this study, competitive benchmarking was included to understand how ergonomic and adjustable functions are represented within the current office chair market. The method was relevant for creating a market-based perspective on which functions are commonly expected in ergonomic office chairs. This supported Research Question 1 by contributing to the understanding of which functions may be considered essential for a shared office chair platform.

### 3.2.3. Questionnaire

Questionnaires are a quantitative method used to collect structured information from a larger group of participants. By using predefined questions, the method allows responses to be compared and analysed in a systematic way [37]. In design research, questionnaires can provide valuable insight into how users interact, behave, what their experience with products are, and which aspects of a product are considered important from the user's perspective [34].

In the context of this study, the questionnaire was included as a method to gather data of which adjustment functions, they consider important in everyday work situations. The purpose of these questions was to identify which functions users consider essential for an ergonomic office chair from a user perspective. This supported Research Question 1 by contributing to the understanding of which functions may be considered essential for an ergonomic office chair within a shared product platform.

### 3.2.4. Observations

Observation is a qualitative research method used to study people, activities and processes in real contexts. By observing users or activities in their natural environment, researchers can gain insight into how products are used in practice and identify behaviours that may not always be captured by asking questions [38]. In design research, observations can be useful because they show how products are used and how people behave in practice, instead of only relying on what participants say in interviews or questionnaires [37].

In this study, observations were included to understand two parts of the design context: how users interact with office chairs, and how office chairs are assembled within a production environment. Together, these perspectives supported the understanding of which functions are important for users and how a shared office chair platform

could be structured. This made the method relevant for both Research Question 1 and 2 as it contributed to the understanding of which functions may be considered and how to design a shared product platform.

### 3.2.5. Interview

Interviews are a qualitative research method used to collect information by asking participants questions about a specific topic. They are commonly used to understand people's experiences, opinions and motivations in more depth [37]. Interviews can have different levels of structure. In structured interviews, the questions follow a fixed order, while unstructured interviews are more open and flexible. Semi-structured interviews are positioned between these two approaches, as they follow a general thematic framework but still allow flexibility in how questions are asked and followed up [39]

In the context of this study semi-structured interviews were included to understand how office chairs are evaluated from a purchasing perspective. This approach was relevant because it allowed key themes, such as chair functionality, product variation and purchasing decisions, to be discussed while still giving space for participants to explain their reasoning. Since purchasers are often involved in selecting office chairs for organizations, their perspective contributed to a broader understanding of requirements beyond the individual user perspective. The method was therefore relevant for both Research Question 1 and Research Question 2.

### 3.2.1.Trend spotting

Trend spotting is an exploratory method often used in design research to identify patterns, changes and directions that may influence future user needs, product expectations and design opportunities. The method is often used to make sense of observed changes in markets, lifestyles and user behaviour by combining different sources of information, such as environmental scanning, trend analysis and pattern recognition. Rather than predicting the future with certainty, trend spotting helps designers and innovators interpret possible directions and reduce the risk of relying only on personal assumptions [40].

In this study, trend spotting was included to understand how office environments and workplace behaviours are changing. This was relevant because the office chair concept needed to respond not only to ergonomic needs, but also to changing expectations of flexibility, aesthetics and use in modern workplaces. The method supported Research Question 2 by contributing to the understanding of how a chair concept could be designed.

## 3.3. Define

The Define phase focuses on analysing and structuring the material collected during the Discover phase. The aim is to identify key insights, clarify the main problem and define the direction for the following design work [31].

### 3.3.1. Personas

Personas were created to represent the target users based on the research conducted in this study. A persona is a research-driven fictional character that represents a realistic user group, including their behaviors, motivations, needs, goals and frustrations [13]. Personas are commonly used in design research to make user insights more concrete and easier to understand during the design process. By connecting user needs to a fictional but realistic person, the method can help designers build empathy and better understand who they are designing for and

why. Personas can also include aspects such as the user's interests, preferences, work situation and products they use, in order to create a more complete picture of the user [35].

In the context of this study, Personas were created to represent the target users based on the research conducted in this study. The personas helped describe different user needs, expectations and challenges connected to ergonomic office chairs and adjustable functions. The purpose of creating personas was to support the design process by keeping the user perspective present when identifying important chair functions and evaluating how a shared product platform could meet different user needs. The personas therefore contributed to understanding of which ergonomic functions are important from a user perspective and what they value, connected to both Research Question 1 and 2.

### 3.3.2. Moodboard

Moodboards are commonly used in design research and product development as a visual method for gathering images, colors, textures, materials, shapes and expressions that represent the intended feeling of a product or brand. The purpose of a moodboard is not to show the final product in a literal way, but rather to communicate the overall atmosphere, design language and qualities that the product should express. Moodboards are often used in the early stages of a design process to explore possible directions, support creativity and make it easier to discuss and evaluate the intended design expression before detailed concept development begins [35].

In the context of this study, the moodboard was used to guide the aesthetic direction of the office chair collection by collecting visual references related to form, color, material and overall expression. The mood board therefore helped create a clearer design direction that supported research question 2.

## 3.4. Develop

The Develop phase focuses on generating, exploring and evaluating possible solutions. During this phase, ideas are developed, tested and compared to move closer to a suitable design proposal [32].

### 3.4.1.Concept selection

Concept selection is a method used in product development to evaluate different alternatives against customer needs and other relevant criteria. The purpose is to compare the strengths and weaknesses of different alternatives in a structured way and to support decisions about which alternatives should be further developed. In the design process, concept selection is often used after idea generation to narrow down several alternatives and decide which ones should be developed further. However, the process can still be iterative, as concepts may be combined, refined or redefined before a final decision is made [25].

Concept selection can be carried out in different ways depending on the purpose of the evaluation. Simpler approaches may involve comparing advantages and disadvantages, while more structured approaches can use evaluation matrices, concept screening or concept scoring. Concept screening is used for a broader comparison between alternatives, while concept scoring uses weighted criteria and a rating scale to support a more detailed evaluation [25].

In this study, the concept selection and scoring method was modified to fit the purpose of the project. Instead of comparing complete product concepts, the method was used to evaluate individual ergonomic adjustment functions. This adaptation was relevant because the aim at this stage was not to select one final chair concept, but to identify which functions should be considered important for an ergonomic office chair based on a shared product platform. As mentioned above, concept selection can be applied at different levels of detail in the development process, including the selection of smaller elements within a larger product concept. This method was there for used to support research question 1.

### 3.4.2. Brainstorming

Brainstorming is a method where ideas are created by temporarily postponing limitations and judgement, allowing the designer to think freely and come up with many different ideas and solutions. During a brainstorming session, ideas are often written down or quickly sketched so they can be remembered and reviewed later. A central part of the method is that criticism should be avoided during the idea generation phase, since the focus is on producing a high quantity of ideas rather than evaluating their quality too early. This helps create a more open and freer mindset where both realistic and more unexpected ideas can be explored. Brainstorming can be carried out in a group, but it can also be done individually, which is often referred to as individual brainstorming [37].

In the context of this study, individual brainstorming was used to explore different directions for the office chair platform, including ergonomic functions, construction principles and possible variations within the chair collection.

### 3.4.3.Mock- up

A mock-up is a rough physical model that is often made full scale and with simple materials such as cardboard, foam or wood. Mock-ups are usually quick and inexpensive to create, and the purpose is not to make a model that looks like the final product, but rather to understand and evaluate an idea in a physical way. In design development, mock-ups can be used to explore aspects such as proportions, form, ergonomics, dimensions and overall expression before more detailed prototypes are developed. Since mock-ups are relatively easy to make, they are often used in the early stages of the design process but can also be used later when specific details or functions need to be tested [35]. In the context of this study, mock-ups were used to investigate how different chair functions could be positioned in relation to the user and the overall chair design. This made it possible to physically explore how controls, mechanisms and design elements could be placed so they were understandable, reachable and integrated into the concept, supporting research question 2.

## 3.5. Deliver

The Deliver phase focuses on refining and presenting the final solution. In this phase, the selected concept is further developed, communicated and prepared as a final design proposal [33].

### 3.5.1. 3D physical model

Prototyping is a method used in design development to give physical form to an idea or concept. Physical prototypes can help communicate and evaluate aspects such as proportions, form, scale, aesthetics and basic functionality. They can also make a concept easier to understand and discuss, since the design is represented in a tangible way rather than only through sketches or digital models. Physical models can have different purposes depending on what needs to be evaluated. A “looks-like” model focuses on representing the visual appearance, proportions and expression of a design, while a “works-like” model is used to demonstrate or test specific functional aspects [37].

In this study, prototyping was included in the Deliver phase to present the final office chair concept in a physical form. The prototype was mainly used as a looks-like model to communicate the overall form, proportions and design direction of the chair. This supported Research Question 2 by showing how an office chair could be designed based on a shared product platform to support product variation.

## 3.6. Ethical considerations

Ethical considerations were considered throughout the project, especially in relation to the questionnaire, observations, interviews and collaboration with Lanab Group AB. The Swedish Research Council explains that research ethics involve balancing the interest of gaining knowledge with the need to protect participants’ integrity and avoid harm or risk of harm [41]. This was relevant for this project since the study included input from users, purchasers and company representatives.

Participation in the questionnaire, observations and interviews was voluntary, and participants were informed about the purpose of the study. Consent was obtained from all participants to use the collected information from the observations and interviews as part of the thesis project. For the interviews, consent was also obtained to record the conversations. The recordings were only used to support transcription and were deleted after the interviews had been transcribed. The material collected was used only for the thesis project, and no personal names or identifying information were presented in the report. During the observations, the focus was either on how users interacted with office chair functions or on understanding the chair assembly and platform-related structure at Lanab Group AB. In both cases, the observations were used to understand the product and design context, not to evaluate individual participants or employees.

Since the project was carried out in collaboration with Lanab Group AB, internal company information was also handled responsibly. Details that could be considered sensitive, such as specific production or business information, were avoided or described on a general level.

# 4. Implementation and result for methods

This chapter presents the implementation of the methods and the results that emerged from the study process.

## 4.1. Discover

### 4.1.1. Literature review

A literature study was conducted to identify which ergonomic chair functions were most relevant for seated office work. The search was carried out through JU Primo Search Library, supported by an AI assistant to identify relevant keywords, articles, and connections between ergonomic functions and body support. The focus was not to conduct a systematic literature review, but to collect literature that could explain why specific functions are important and how they contribute to posture, comfort, movement, and pressure distribution.

The search process included both broader and function-specific search phrases. For example, broader searches such as “important adjustable functions in seating ergonomics” were used to identify relevant ergonomic functions in office chairs. More specific searches, such as “why is adjustable backrest height important in seating ergonomics”, were then used to find literature that explained the importance of individual functions. This helped identify sources that did not only mention a function but also explained its ergonomic relevance.

The literature study focused on functions commonly found in adjustable office chairs, including seat height, seat depth, seat angle, backrest height, lumbar support, armrests, lockable backrest tilt, and rocking function. For each function, the sources were reviewed to see whether they supported the function as ergonomically relevant and whether they explained its effect on the body. The information gathered was compiled in Chapter 2, *Theoretical Framework,* under the section *Seating Ergonomics*. In total, 27 sources were included in the seating ergonomics section, where 19 sources directly supported one or more ergonomic chair functions.

The table below summarises how many references were identified for each function. The number should not be interpreted as a statistical ranking, since parts of the literature search were function specific. Instead, it gives an overview of the amount of literature support identified for each function and was later used as one input in the concept selection process.

***Table 1:*** *literature support levels based on the number of supporting references identified for each ergonomic function.*

| Literature support level | Number of supporting references |
|---|---|
| Very strong | 7 or more references |
| Strong | 5–6 references |
| Moderate/Strong | 4 references |
| Moderate | 2–3 references |
| Low | 1 reference |
| No clear support | 0 references |

*Table 2: Overview of adjustable ergonomic chair functions, including the number of supporting references, literature support level and identified references.*

| Ergonomic function | Number of supporting references | Literature support level | Supporting references |
|---|---|---|---|
| Seat height adjustment | 9 | Very strong | [11], [12], [13], [15], [16], [17], [18], [19], [20] |
| Seat depth adjustment | 7 | Very strong | [11], [12], [15], [17], [18], [19], [20] |
| Seat angle adjustment | 6 | Strong | [11], [12], [13], [15], [16], [19] |
| Backrest height adjustment | 11 | Very strong | [11], [12], [13], [17], [19], [20], [21], [22], [23], [24], [25] |
| Lumbar support | 5 | Strong | [10], [11], [12], [13], [25] |
| Armrest adjustment | 7 | Very strong | [3], [10], [12], [13], [16], [17], [18] |
| Lockable backrest tilt | 7 | Very strong | [3], [10], [11], [12], [13], [17], [19] |
| Rocking function | 7 | Very strong | [7], [11], [12], [13], [23], [26], [27] |
| Tilt tension Control | 1 | Low | [28] |

### 4.1.2. Competitive Benchmarking analysis

To investigate the ergonomic features provided by leading office chair companies, a competitive benchmark analysis was conducted. Eight leading companies within the office seating market were selected: Malmstolen, Flokk, Herman Miller, Kinnarps, Humanscale, IKEA, Interstuhl and SAVO. For each brand, four to five popular or best-selling chairs were reviewed by collecting product information through product specification sheets and product demonstration videos. Based on this material, the available adjustable functions for each chair were documented. The data collected was organized into a competitive benchmarking matrix using Microsoft software Excel. In this matrix, ergonomic chair functions were listed along the horizontal axis, while the vertical axis

contained the selected companies and their corresponding chair models. Each chair model was evaluated based on whether a specific function was present or not. The matrix is presented in appendix A.

After all ergonomic chair models had been reviewed and documented, the total number of chairs offering each function was calculated. To show how regularly various functions appeared across the reviewed chairs, the frequency of each function was converted into percentages. The results were visualized in a bar to simplify the results *figure 10.*

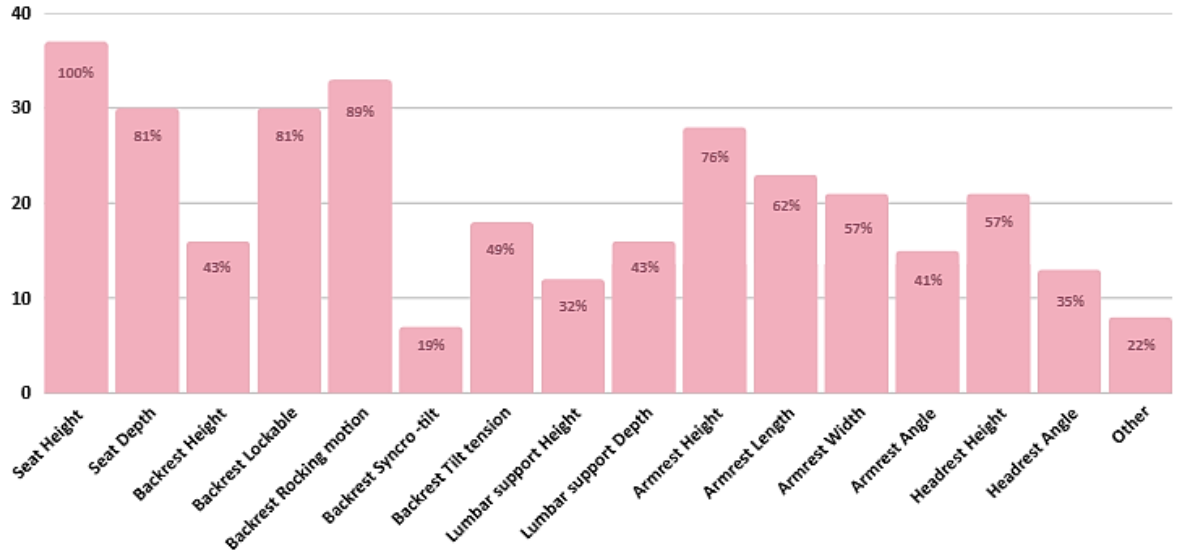



*Figure 10: Frequency of ergonomic adjustment functions across analysed office chair models of the competitors.*

The results showed that several ergonomic adjustments were present in a large majority of the chairs reviewed. Seat height adjustment was present in every reviewed model, and seat depth adjustment, backrest locking, backrest rocking motion and armrest adjustments (height, length) were also highly present in reviewed chairs. Other adjustment functions appeared less frequently such lumbar support depth, Synchro- tilt, headrest angle, and other functions.

### 4.1.3. Questionnaire

To gather user insight regarding their office chairs and their function, a questionnaire was conducted. The questionnaire was made and published by Microsoft Forms and shared with approximately 400 of Lanab Group Ab´s customers as well as through professional networks on social platforms such as LinkedIn. 82 responses were collected in total and used for analysis.

The questionnaire was structured around three main questions related to office chair functions. Participants were asked which adjustable functions their current office chair has, which of these functions they use regularly, and which functions they consider necessary if they buy a new office chair. These three perspectives were used to identify differences between the availability of functions, their actual use, and their perceived importance. The

The collected responses were first summarised in a bar chart presented as *figure 11 to* visualise the distribution of the answers across the three categories: *has*, *uses*, and *must*. The data was then added into a table with the

data converted to the percentage of participants who reported the functions they have, use, and must have. Based on these responds, a prioritisation column was also added in the table to categorise the functions according to their relative importance among participants.

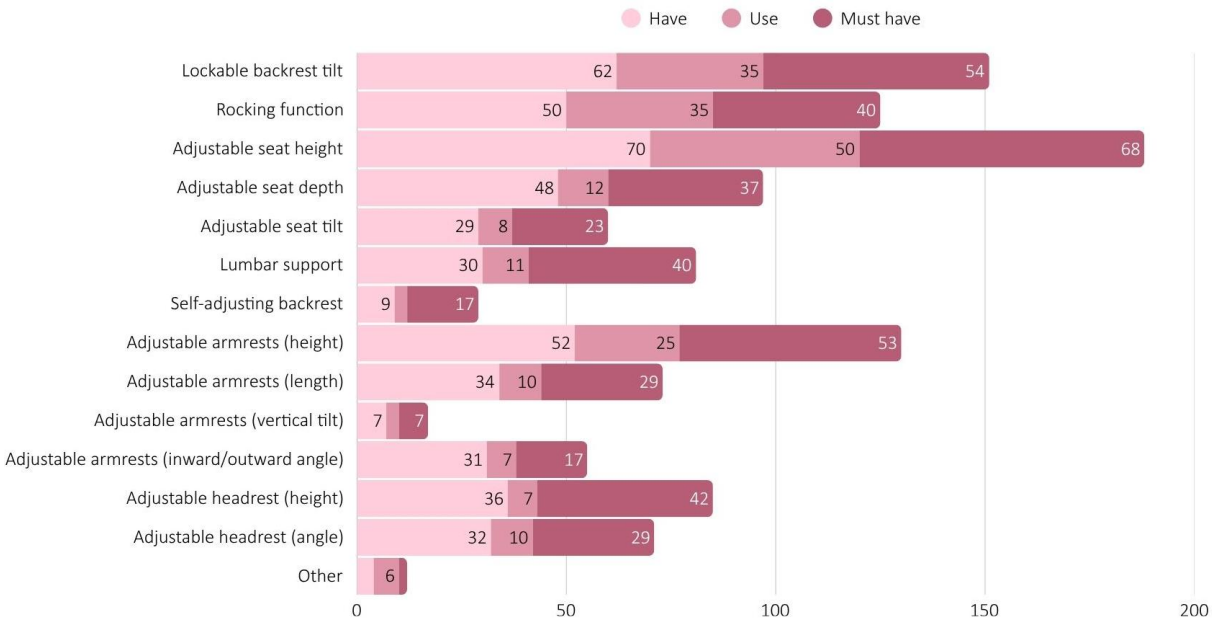



***Figure 11:*** *Questionnaire results showing which adjustable chair functions participants have, use and consider necessary.*

***Table 3:*** *Questionnaire results presented as percentages for each adjustable chair function, where n represents the number of responses for each question.*

| | Has (%) (n=76) | Use (%) (n=74) | Must have (%) (n=72) |
|---|---|---|---|
| Backrest height | 76% | 18% | 58% |
| Lockable backrest tilt | 81% | 47% | 75% |
| Rocking function | 66% | 47% | 56% |
| Seat height | 92% | 68% | 94% |
| Seat depth | 63% | 16% | 51% |
| Seat angle | 38% | 11% | 32% |
| Lumbar support | 41% | 15% | 56% |
| Self-adjusting backrest | 12% | 4% | 24% |
| Armrests (height) | 68% | 34% | 74% |
| Armrests (length) | 45% | 14% | 40% |

| | | | |
|---|---|---|---|
| Armrests (vertical tilt) | 9% | 4% | 10% |
| Armrests (inward/outward angle) | 41% | 9% | 24% |
| Headrest (height) | 47% | 9% | 59% |
| Headrest (angle) | 42% | 14% | 40% |
| Other | 5% | 14% | 3% |

The results showed that adjustable seat height was the function that users mostly reported their chairs having, with 92% of the respondents indicating having it. Lockable backrest tilt and adjustable armrests height were also commonly reported, appearing in 81% and 68% of the reported chairs respectively. When it came to the functions the respondent actively uses during their workday, adjustable seat height adjustment was also the most frequently used (68%). Other commonly used features among the respondent were lockable backrest tilt (47%), rocking function (47%) and armrest height (34%). In contrast, functions such as adjustable seat depth, adjustable armrest angle, and adjustable seat tilt were used less frequently among the respondents.

Respondents reported that an adjustable seat was essential for them to consider buying an office chair, with 94% of respondents indicating it as a requirement. Adjustable armrests height (74%) and lockable backrest tilt (75%) were also indicated essential by many respondents. Other functions such as lumbar support and adjustable headrest height were also identified as important by a notable share of participants. In contrast, Adjustable armrests with both vertical tilt and inward/outward angle, as well as adjustable seat tilt, were not reported as highly required functions among respondents.

### 4.1.4. Observation

#### 4.1.4.1. User Observation.

Ten participants who work in private office environments and regularly spend extended periods sitting during their workday were selected for the user observation. The participants were recruited through an existing contact and further recommendations, which made it possible to reach people of different ages and genders who were currently working in office-based roles. Each potential participant was contacted privately and asked whether they usually worked in an office environment for approximately eight hours or more, and whether they had an office setup where the observation could be carried out. Based on this, ten suitable participants were selected. The participants included both men and women between the ages of 20 and 60. The observations took place in the participants' own office environments to capture natural interactions with office chair adjustments during normal work activities.

During the observation, three office chair models from Lanab Group AB were used: Surf Synkron, Höganäs 280, and the LD 6340. The chairs included several adjustable ergonomic functions, such as seat height, seat depth, backrest height, lockable backrest tilt, synchro tilt, rocking function and tilt tension control. Depending on the chair model, additional functions such as adjustable lumbar support, seat angle, armrests and headrest were also available. Each session began with the participant adjusting the Höganäs chair at their desk without receiving any

instructions or guidance. All chairs were reset to a default configuration before each observation, meaning that they were adjusted to the same initial settings, such as the lowest seat height and a neutral backrest position.

Participants were encouraged to think aloud while interacting with the chair and to express their thoughts during the adjustment process. Their verbal comments were documented together with observed behaviours and interactions with the chair mechanisms. Notes were taken regarding which functions participants attempted to adjust, how they interacted with the controls, and how they responded to different adjustment mechanisms. After the participant had adjusted the chair to a comfortable working position, they were asked the question: “Is there any function that you expected to find but could not locate?” The participant then continued working in the chair for approximately one hour to observe how the chair functions were used during normal work activities. After this period, the participant was asked whether any additional adjustments had been made while working. The same procedure was then repeated for the remaining two chair models. In total, each observation session lasted approximately three hours.



***Figure 12:*** *The three office chairs that was used in the observation. The left one is Surf Synkron, middle is LD 6340, and the chair on the right side is Höganäs 280.*

#### 4.1.3.1.1 Results of behavioral interaction

The observations revealed five repetitive behaviour patterns in how participants interacted with all the three models These patterns primarily related to user expectations, the clarity of control mechanisms, and the ability to identify and understand different functions.

##### User expectations

A clear pattern observed during the sessions was that participants had strong expectations regarding where certain adjustment functions should be located. Many users instinctively searched for seat height, lockable backrest, and backrest height controls on the right side of the chair. When these controls were not located where expected, participants often began searching around the chair or attempted to reach underneath the seat. In

some cases, participants had to stand up or change posture to locate the controls. When the expected functions were not found immediately, participants expressed frustration or uncertainty.

#### Similar controls creating confusion

Another recurring observation was that chairs with multiple similar looking buttons positioned close to each other created uncertainty for the participants. When multiple controls were positioned close to each other or had similar shapes, participants sometimes struggled to identify which button controlled which function. Several participants needed to test different controls to understand their purpose. In some cases, participants stated that they forgot the purpose of certain buttons during the session and needed to search for them again when attempting to make additional adjustments.

#### Importance of visual guidance

The observations also showed that visual guidance played an important role in helping users understand chair functions. When participants discovered visual indicators such as icons and graphical instructions on the buttons at the LD chair, their understanding increased and participants were generally able to understand the function more quickly. In contrast, when the controls lacked visual indicators, participants often relied on trial-and-error interactions to understand how the mechanism worked.

#### Difficulty locating certain adjustments

Some adjustment functions were consistently more difficult for participants to locate. In particular, the backrest height adjustment was frequently difficult for users to find. Several participants searched around the chair or attempted multiple interactions before identifying the correct control. In some cases, participants adjusted their posture, bent down, or stood up in order to locate the mechanism. A few participants stopped searching after some time and continued using the chair without adjusting the function.

#### Unclear operational logic of certain mechanisms

Participants also expressed uncertainty regarding how certain mechanisms were intended to be operated. Some adjustment controls required specific actions such as pulling, pushing, or rotating, but the expected behaviour was not always clear to the users. In particular, mechanisms related to rotation-based tension adjustments and pump-like controls caused confusion among several participants. Some participants expressed uncertainty about whether the adjustment had actually been activated, partly due to the lack of clear feedback from the mechanism.

##### 4.1.3.1.2 Result of Function usage

The observation sessions also recorded which chair functions participants adjusted during the interaction. Seat height adjustment was used by all participants, while backrest tilt lock and armrest height adjustment were also used by a majority of the participants. Several other functions, such as seat depth adjustment, headrest adjustments, and lumbar support depth, were used less frequently during the sessions.

***Table 4:*** *Observation results showing the number and percentage of participants who used each adjustable chair function during the observation sessions.*

| | Count n= 30 | Percentage (%) |
|---|---|---|
| Seat hight | 30 | 100% |
| Seat angle | 0 | 0% |
| Seat depth | 14 | 47% |
| Backrest height | 19 | 63% |
| Lockable backrest tilt | 23 | 77% |
| Rocking motion | 6 | 20% |
| Tilt Tension Control | 3 | 10% |
| Armrest (Height) | 19 | 63% |
| Armrest (Width) | 10 | 33% |
| Headrest (height) | 14 | 47% |
| Headrest (angle) | 8 | 60% |
| Lumbar support depth | 4 | 13% |

### 4.1.5.Assembler observation

To understand how office chairs are assembled and how different components are connected, an assembly observation was conducted at Lanab Group AB. The observation focused on identifying assembly structures, component dependencies, and how ergonomic functions were integrated within different chair models. The observation was carried out by following an assembler through the assembly process, from receiving the order to packaging the finished chair. Several steps in the process were observed for three office chair models: Höganäs 280, Surf Synk, and LD 63.



***Figure 13:*** *Assembly observation at Lanab Group AB, showing selected steps from chair assembly and packaging.*

The observation showed that many components arrived as pre-assembled subassemblies from suppliers, while the final assembly was carried out internally at Lanab. Although the observed chair models differed in form, mechanism type and configuration, the overall assembly structure followed a similar pattern. The base, gas lift, seat, backrest and armrests were all connected directly or indirectly to the mechanism, showing that the mechanism functioned as the central hub of the chair structure. From a product platform perspective, this indicated that the mechanism could act as the main platform base, since several ergonomic functions were integrated into or controlled through it, including seat height, seat depth, backrest tilt, rocking function, tilt tension and seat angle. These functions were therefore understood as core platform functions that could remain consistent across different chair variations.

The observation also showed that some components and functions could be added, adapted or varied around the shared mechanism. Armrests, headrests and lumbar support were identified as additional components that

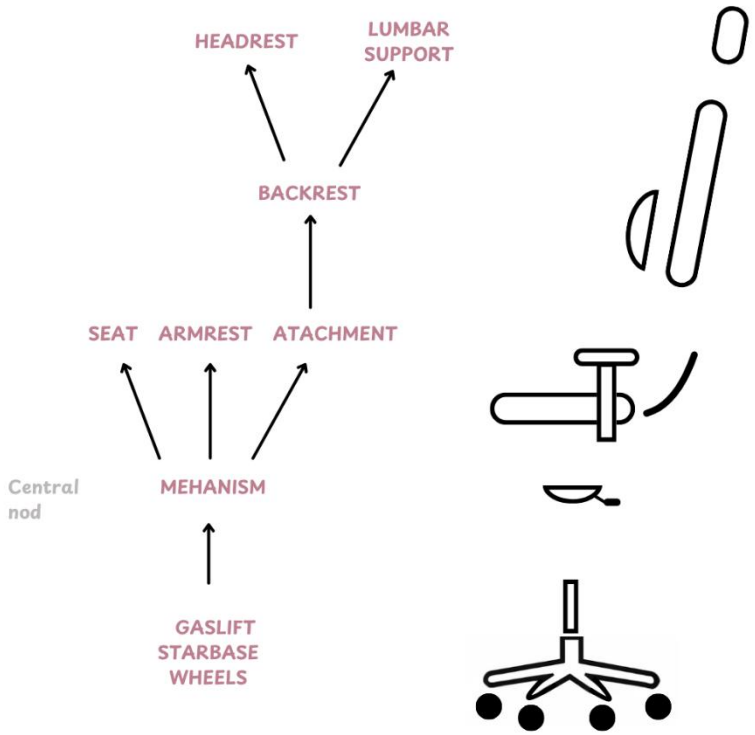



***Figure 14:*** *The relationship between the office chair component and different chair functions observed.*

could be included depending on user needs, purchaser requirements or product version. Backrest height adjustment was connected to the backrest attachment rather than the main mechanism, meaning that it depended more on the specific backrest construction and design. In addition, visible components such as the seat and backrest could vary in shape, material and construction, while upholstery, fabric colour and foam could be selected depending on customer preference. This showed that the platform does not need to define the full visual design of the chair but can instead provide a shared technical structure that supports variation in the surrounding components.

### 4.1.6. Interview with purchaser

To understand the procurement perspective within the office chair industry, semi-structured interviews were conducted with three large public organizations in Sweden: Adda, Domstolsverket and Försäkringskassan. The interviews were held digitally through Microsoft Teams and followed a structured interview format. Each interview began with a short introduction of the project and its purpose, followed by questions regarding how the organizations work with procurement of office chairs, how purchasing processes are structured, and which requirements are considered most important during evaluation and selection. The interviews also explored topics related to ergonomics, standardization, certifications, durability, and long-term use within public work environments.

#### Predefined requirements and framework agreements

A main finding was that public procurement of office chairs is highly structured and strongly connected to predefined requirements, standards and framework agreements. The respondents explained that office chairs are not only evaluated based on comfort, appearance or individual functions, but also on their ability to meet European standards, ergonomic requirements, sustainability criteria and durability expectations. Certifications such as Möbelfakta were described as important, since they include requirements for both the complete product and its materials, such as textiles, plastics and metals. The interview with Försäkringskassan also showed that many public organizations do not define all technical requirements themselves, but instead purchase through national framework agreements, such as Kammarkollegiet's agreement for work chairs. This means that several requirements are already established before the individual organization makes a purchase.

#### Changing Work Environments and Broader User Needs

Another important finding from the Adda interview was that workplace needs have changed over time. The respondent explained that the office chair no longer has the same central role as before, partly because electrically adjustable sit-stand desks have become common in many workplaces. Earlier, when desks were often fixed in height, the office chair needed to provide a larger range of adjustments to compensate for different users and working positions. Today, other ergonomic solutions are integrated into the workplace, which means that not every chair needs to include the most advanced adjustment functions for most users. However, users with specific physical needs may still require more individually adapted chairs.

The interviews also showed that shared work environments have affected how office chairs are evaluated. Activity-based offices, shared desks and shift-based workplaces mean that many chairs are no longer used by one specific person. Instead, several users may share the same chair during a day or week. This creates a need for chairs that are easy to adjust, understandable to use and suitable for a broad range of body types. For the general workplace user, the focus therefore seems to be less on highly specialized individual adjustment and more on standardized ergonomic functionality that can support many different users.

### Design Selection Influenced by Interior Architects

Finally, the interview with Domstolsverket also highlighted that the final selection of office chairs is not only based on ergonomic and technical requirements. Aesthetic and spatial aspects are also important, especially in workplaces where the interior environment needs to follow a certain architectural concept or design direction. In these cases, architects or interior architects can have a strong influence on the final choice of chair, including colour, material, form and how the chair fits into the overall interior. This shows that an office chair must fulfil both functional and visual requirements to be relevant for public-sector environments.

## 4.1.7. Trend spotting

Trend spotting was conducted to understand how the office environment is developing and how these changes could influence the design of a new office chair collection. The work was based on reading trend reports, articles, statistics related to work life and office behaviour, and by studying company webpages to understand how new office furniture and workplace solutions are presented.  The sources were compared to identify repeated patterns rather than single examples. In addition to reports and digital sources, observational studies were also conducted during Milan Design Week 2025 where furniture exhibitions and showroom environments were analysed to identify recurring patterns in form language, materials, colours and spatial aesthetics. Photographs and visual references were documented and later compared with findings from reports and industry publications to identify consistent trends across both conceptual forecasting and commercial design practice. The purpose was not to predict one specific future, but to understand current directions within workplace design and use them as input for the design process. The result of the trend spotting are presented below:

### The office as a social and purpose-driven place

A repeated trend throughout the analysed material was that the office is no longer only viewed as a place for individual desk work. Several reports describe the office as an increasingly social environment focused on collaboration, culture, creativity and interaction between employees [42] [43] [44]. As hybrid work has become more common, the office is increasingly used for activities that are difficult to replicate remotely, such as workshops, informal communication, teamwork and relationship building [45] [44]. At the same time, offices are becoming more purpose-driven, where different zones are designed for different activities rather than one general workspace. Focus rooms, quiet areas, lounges, hybrid meeting rooms and collaborative spaces are repeatedly highlighted throughout the analysed sources [42] [45] [46] [43]. This shows a shift towards more varied office environments where furniture needs to function across multiple types of spaces and interactions.

### Hybrid work and integrated technology

Another clear trend is that hybrid work has become a permanent part of modern working life. Several sources describe how offices increasingly need to support both physical and digital collaboration through hybrid meeting rooms, flexible workstations and integrated technology [45] [47] [46] [44]. Digital tools such as booking systems, video conferencing solutions, smart office technology, and AI are becoming natural parts of the workplace rather than separate additions [47] [46]. The reports also show that work environments are becoming more flexible and adaptable as employees move between different locations and ways of working [45] [44]. This means that office

furniture should not compete with technology or add complexity but support changing work situations in a simple and intuitive way. Products need to be easy to understand, flexible in use and suitable for shared or changing work environments.

### Focus, privacy and recovery from open offices

Several reports also question the fully open office landscape and show an increasing demand for quieter and more private work environments [42] [48] . Focus rooms, phone booths, small meeting rooms and quiet zones are highlighted as important additions to support concentration and reduce distractions [42] [43]. The analysed material shows that employees still value environments that support deep work and mental recovery, even within activity-based workplaces. This indicates a growing need to balance social interaction and collaboration with privacy, calm environments and acoustic comfort.

### Ergonomics, movement and well-being

Health and well-being were recurring themes throughout all analysed material. Ergonomics is no longer only presented as a technical requirement, but as part of a broader focus on employee well-being, movement and long-term health [42] [47]. Several sources describe how office environments increasingly include spaces and solutions that encourage movement, recovery and physical comfort during the workday [42] [44]. Well-being is also connected to mental health, where calmer environments, natural materials and more comfortable workplaces are highlighted as important aspects of future office design [47] [46]. This shows that workplace design is becoming increasingly human-centred, where comfort, movement and adaptability are treated as fundamental parts of the work environment.

### Softer and more human-centred design

A strong visual trend identified throughout both the reports, and the exhibition observations were a shift towards softer and more human-centred office environments. Several sources describe how workplace interiors are becoming increasingly inspired by hospitality and residential environments rather than traditional corporate offices [47] [46] [48] . Lounge-inspired spaces, softer seating, warm atmospheres and more welcoming environments are becoming increasingly common [47] [46]. During Milan Design Week 2025, recurring patterns could also be observed in the furniture industry, particularly regarding rounded geometries, softer transitions, tactile materials and warm colour palettes. Natural materials such as wood, metal and textiles were frequently combined with warm neutral colours, creating calmer and more emotionally supportive environments. This indicates a broader shift towards office environments that feel more approachable, comfortable and socially inviting.

### Sustainability as an emotional value

Sustainability was another recurring trend identified throughout the analysed material. The reports increasingly communicate long-lasting products, responsible material choices and adaptable environments that can remain relevant over time [45] [47] [48] . However, sustainability is not only presented through technical aspects such as durability or material performance, but also through a visual connection to nature. Several sources show a shift towards calmer and more natural office environments, where earthy colour palettes, tactile materials,

wood, textiles and organic forms are used to create a more grounded and human-centred atmosphere [47] [46] [48] . Rather than focusing on short-term visual trends, many companies present durability, timeless aesthetics and flexibility as central values within workplace design [47] [46]. Adaptability is also described as part of sustainability, where flexible spaces and products can evolve together with changing workplace needs instead of being replaced entirely [45] [48] . This shows a shift towards more long-term and resilient approaches to office environments and furniture design, where sustainability is expressed both through function and through a natural, calming design language.

## 4.2. Define

### 4.2.1.Personas

Three personas were created to represent the different stakeholders (end user, purchaser and architect) identified during the project and to make their needs easier to understand in the design process. The personas were based on insights gathered from interviews, the questionnaire and observations, which were combined to create three fictional characters. Each persona represented a different perspective connected to the office chair platform, including user needs, purchasing needs and company-related needs. By translating the collected data into personas, the findings became more concrete and easier to relate to during the development of the




***Figure 15:*** *Persona representing the end user perspective.*




***Figure 16:*** *Persona representing the purchaser perspective.*




***Figure 17:*** *Persona representing the architect perspective.*

### 4.2.2.Design direction & Moodboard

A mood board was created to translate the research findings into a clearer visual direction for the design development. The design direction was based on the user needs identified through the research, together with the trends found in the trend analysis. From this, three keywords were selected to guide the design: minimal,

**MINIMAL**
Focus on essential functions. Clean, clear, and simple design.

**INTUITIVE**
Functions and adjustments should be easy to understand and use without instructions.

**SOFT**
Soft shapes and smooth transitions. A welcoming and comfortable expression.

intuitive and soft. These keywords were chosen to support a chair concept that feels simple, easy to understand and welcoming in different office environments.

The moodboard was then created with 5 images collected from Pinterest, that represent the three keywords. The minimal direction focuses on reducing visual complexity and keeping the design clean and clear, with attention to essential functions rather than unnecessary details. The intuitive direction focuses on how adjustments and functions should be easy to understand and use without instructions. The soft direction relates where rounded shapes, smooth transitions and comfortable materials create a more approachable and human feeling.




***Figure 18:*** Moodboard representing the tree keywords.

A colour palette was also chosen as part of the design direction. The colours were based on the trend analysis, which highlighted a movement towards warmer, more natural and flexible office environments. The palette was therefore selected to reflect Scandinavian nature and sustainability, while also creating a welcoming feeling in modern work settings. Green represents nature, calmness and relaxation, while brown adds warmth, comfort and a more natural feeling. Orange brings energy, enthusiasm and positivity, and blue represents productivity,

stability and calm. These colors were mainly intended for textile variations, while the base colors were kept more neutral to create a flexible platform that can work across different chair versions and office settings.

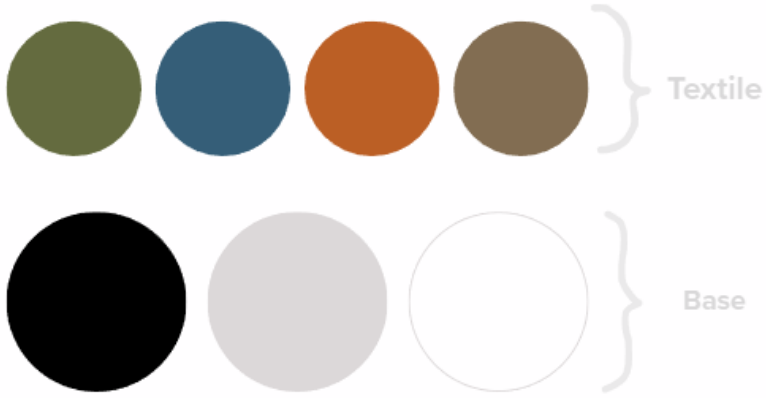



***Figure 19**: Selected colour palette for the chair concept, including textile colours and base colours.*

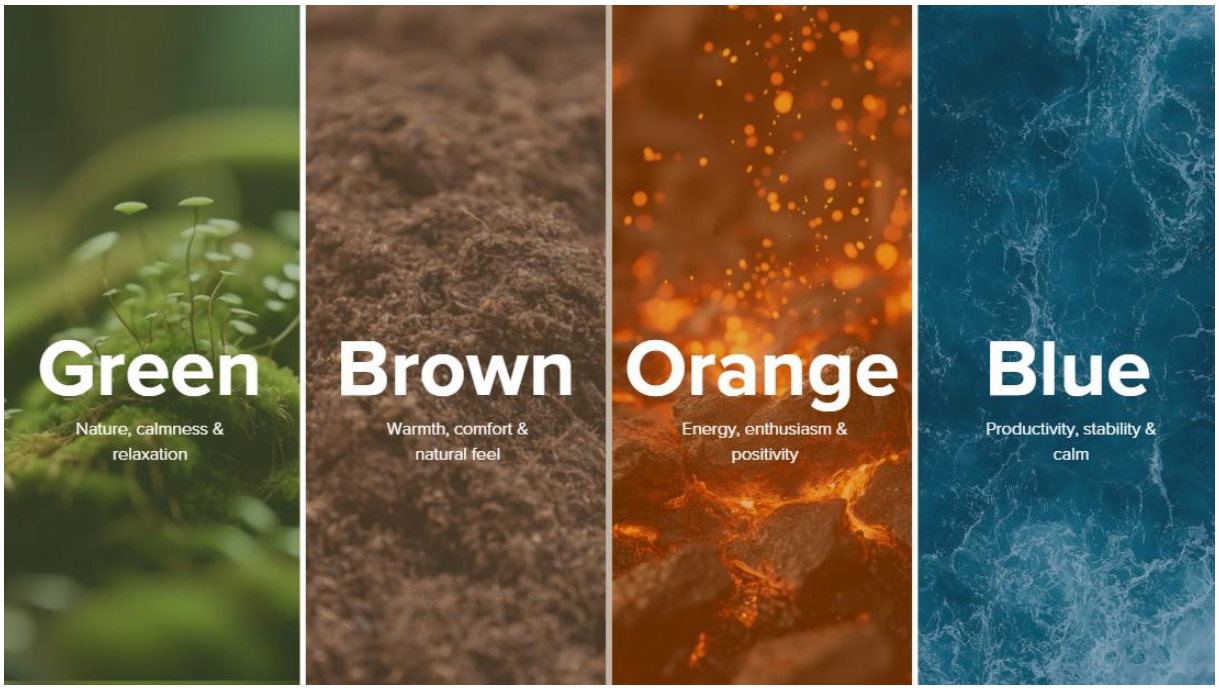



***Figure 20:**. Colour inspiration for the textile palette, showing the intended associations behind each selected colour.*

## 4.3. Develop

### 4.3.1. Concept selection

Concept selection was conducted to decide the relevant function to include in the platform. All collected data regarding the ergonomic adjustment functions was summarized in a evaluation matrix. The matrix included the

survey results, such as how many respondents stated each function a must-have. It also included the observed use of functions during the user observations, the frequency of each function in the competitive market benchmarking, and the ergonomic relevance identified in the literature review. By collecting these different sources in one matrix, it became easier to compare each function from several perspectives: what users say they need, what they use, what is common on the market, and what is supported by ergonomic theory. This created a clearer basis for deciding which functions were most relevant for the shared platform and the matrix is presented as table 5.

***Table 5:*** *Evaluation matrix for adjustable chair functions based on collected project data.*

| Function | Must Have (%) | Use (%) | Competitor Presence (%) | Literature Support |
|---|---|---|---|---|
| Seat height | 94% | 68% | 100% | Very strong |
| Lockable backrest tilt | 75% | 47% | 81% | Very strong |
| Rocking function | 56% | 47% | 89% | Very strong |
| Backrest height | 58% | 18% | 43% | Very strong |
| Seat depth | 51% | 16% | 81% | Very strong |
| Seat angle | 32% | 11% | — | Strong |
| Lumbar support | 56% | 15% | 32% | Strong |
| Armrests (height) | 74% | 34% | 76% | Strong |
| Armrests (length) | 40% | 14% | 62% | Moderate |
| Armrests (inward/outward angle) | 24% | 9% | 41% | Moderate |
| Headrest (height) | 59% | 9% | 57% | — |
| Headrest (angle) | 40% | 14% | 35% | — |
| Tilt tension control | — | — | 49% | Low |

After the data had been summarized, a modified scoring method was used to make the evaluation more structured. Each function was rated from 1 to 5 for the selected criteria, where a higher rating indicated stronger support. The criteria were then weighted depending on how important they were for the aim of the project. For example, user importance and literature support were given higher weight because the project focused on

identifying essential ergonomic functions, while platform compatibility was included to evaluate whether the function could fit within a shared platform structure.

***Table 6:*** *Rating scale used to convert the collected data for each criterion into ratings.*

| Rating | Must have % | Use % | Competitor presence % | Literature support |
|---|---|---|---|---|
| 5 | ≥80% | ≥60% | ≥80% | Very strong |
| 4 | 60–79% | 40–59% | 60–79% | Strong |
| 3 | 40–59% | 15–39% | 40–59% | Moderate |
| 2 | 20–39% | 5–14% | 20–39% | Low |
| 1 | <20% | <5% | <20% / no data | — / No data |

***Table 7:*** *Weighting of the selected criteria.*

| Criteria | Weight |
|---|---|
| Must have | 30% |
| Use | 30% |
| Competitor presence | 15% |
| Literature support | 25% |
| **Total** | **100%** |

After the criteria and weights had been defined, each function was rated against the selected criteria. The rating was then multiplied by the weight of each criterion and calculate a weighted score:

**Weighted score = *rating* × *weight***

The weighted scores were then added together to create one final score for each function. This allowed the functions to be compared in a more systematic way, since the evaluation considered several sources of data rather than one factor alone.

***Table 8:*** *Final scoring matrix showing the rating, final score and platform decision for each adjustable chair function.*

| Function | Must have rating | Use rating | Competitor rating | Literature rating | Final score | Decision |
|---|---|---|---|---|---|---|
| Seat height | 5 | 5 | 5 | 5 | **5.00** | **Core platform** |
| Lockable backrest tilt | 4 | 4 | 5 | 5 | **4.40** | **Core platform** |
| Rocking function | 3 | 4 | 5 | 5 | **4.10** | **Core platform** |
| Backrest height | 3 | 3 | 3 | 5 | **3.50** | **Platform support** |
| Seat depth | 3 | 3 | 5 | 5 | **3.80** | **Core platform** |
| Seat angle | 2 | 2 | 1 | 4 | **2.35** | **No go/exclude** |
| Lumbar support | 3 | 3 | 2 | 4 | **3.10** | **Addable function** |
| Armrests height | 4 | 3 | 4 | 4 | **3.70** | **Addable function** |
| Armrests length | 3 | 2 | 4 | 3 | **2.85** | **No go/exclude** |
| Armrests inward/outward angle | 2 | 2 | 3 | 3 | **2.40** | **No go/exclude** |
| Headrest height | 3 | 2 | 3 | 1 | **2.20** | **Addable function** |
| Headrest angle | 3 | 2 | 2 | 1 | **2.05** | **Addable function** |
| Tilt tension control | 1 | 1 | 3 | 2 | **1.55** | **No go/exclude** |

The final scoring matrix shows that adjustable seat height, seat depth, lockable backrest tilt and rocking function received the highest scores and were therefore considered most suitable to include in the shared platform these functions are directly connected to the mechani sm and form the shared ergonomic base of the chair. Backrest height also received a high score and was included in the platform, but not as part of the core mechanism. This decision was based on the assembly observation, which showed that the backrest is connected to the mechanism through a separate attachment. Armrest and headrest functions were not included in the core platform but were instead considered addable functions. This means that they can be added depending on user needs, purchaser requirements or chair variant, without being part of the shared platform base. Tilt tension control and seat angle received one of the lowest scores and was therefore not prioritised for the core platform.

### 4.3.2. Brainstorming

Brainstorming was used throughout the concept development phase as a method to explore, visualize and refine different design possibilities for the office chair platform. The process was carried out in several sessions individually, where each session focused on a specific part of the chair design. Sketching was used as the main

tool during these sessions, since it made it possible to quickly generate many ideas and compare different directions before developing them further.

The first brainstorming sessions focused mainly on form exploration. Different backrest shapes were sketched individually to investigate how the chair could express a soft design language. Simple 2D side views were also created to study the overall silhouette of the chair and to understand how the seat, backrest and base could relate to each other visually. This helped define the early design direction and supported decisions about the chair's proportions and visual balance.

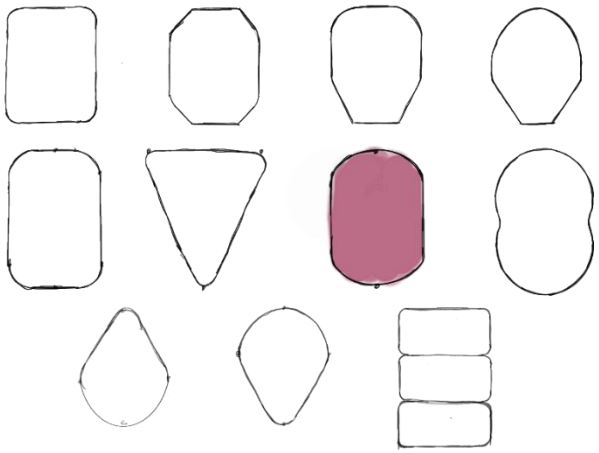


***Figure 21:*** *Sketches of Backrest shape exploration during the early design process.*

The exploration then continued with the attachment between the seat and the backrest. Different connection solutions and support structures were explored through sketches. The backrest attachment was sketched from the side views to see how it could look like when it connects the backrest to the mechanism.

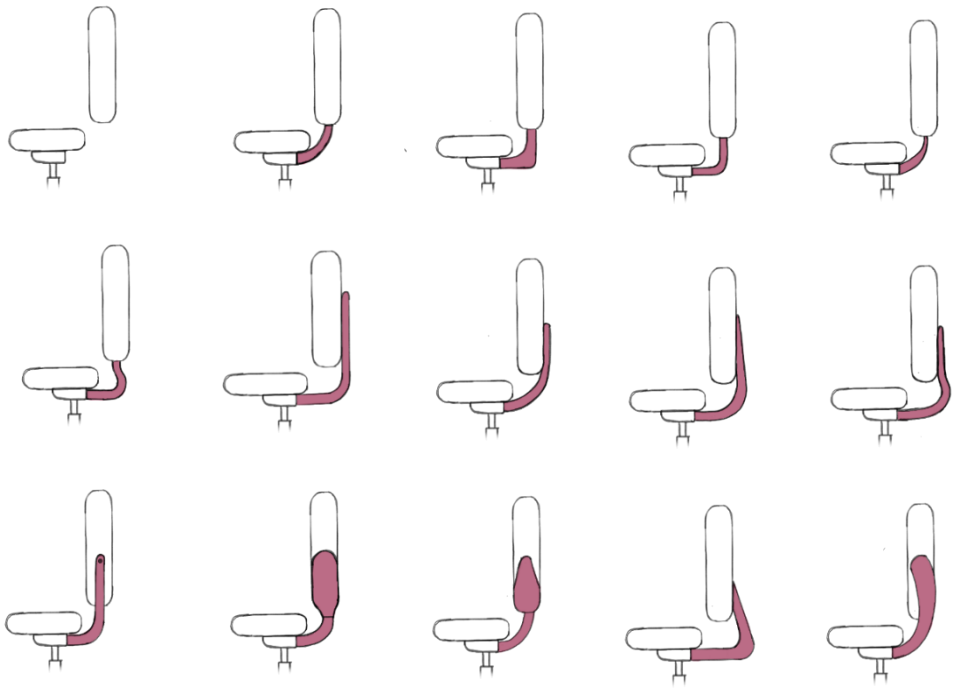


***Figure 22:*** *Brainstorming sketches of backrest attachment from sideview.*

Based on the shape and attachment explorations, four concept directions were developed using digital sketching tools such as Procreate. Each concept represented a different interpretation of the design language and platform structure. The concepts were then compared and discussed with guidance from Lanab Group. Through this process, concept one was selected for further development because of its simple and modern expression, as well as its potential to be integrated into the product platform without requiring overly complex construction solutions. This made the concept suitable from both a design and implementation perspective.

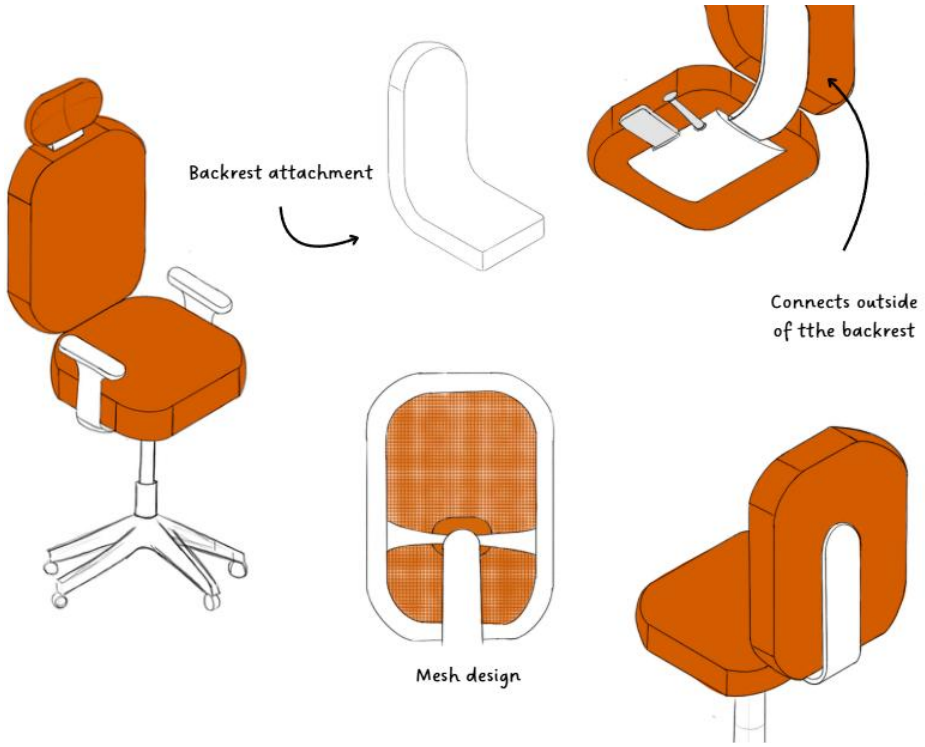



***Figure 23:*** *Digital sketches of Concept 1.*

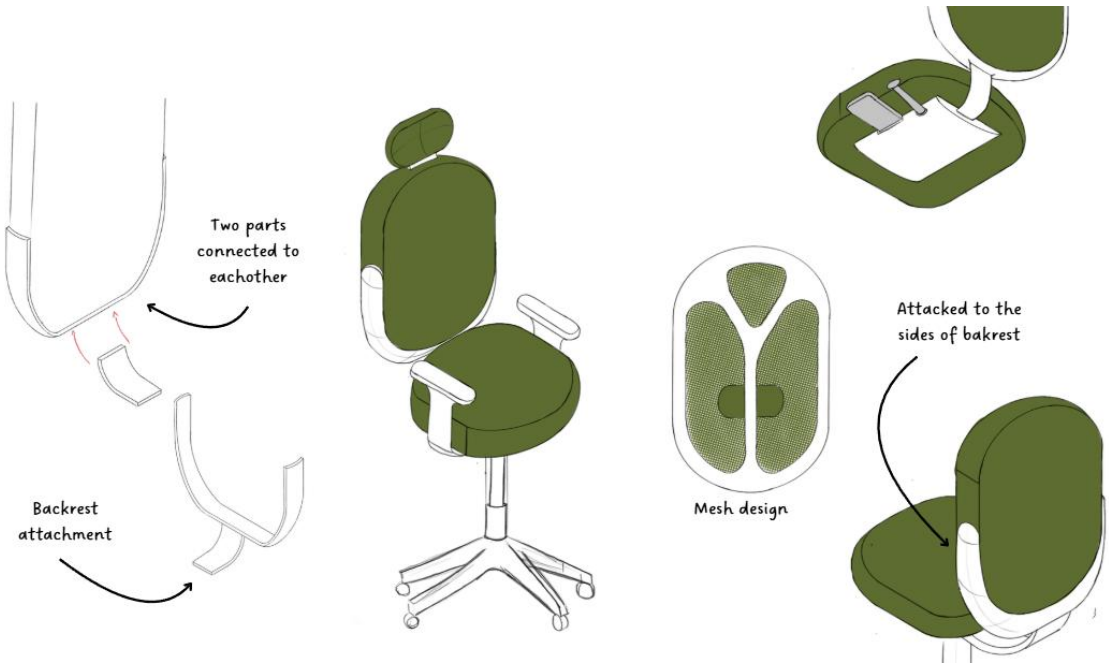



***Figure 24:*** *Digital sketches of Concept 2.*

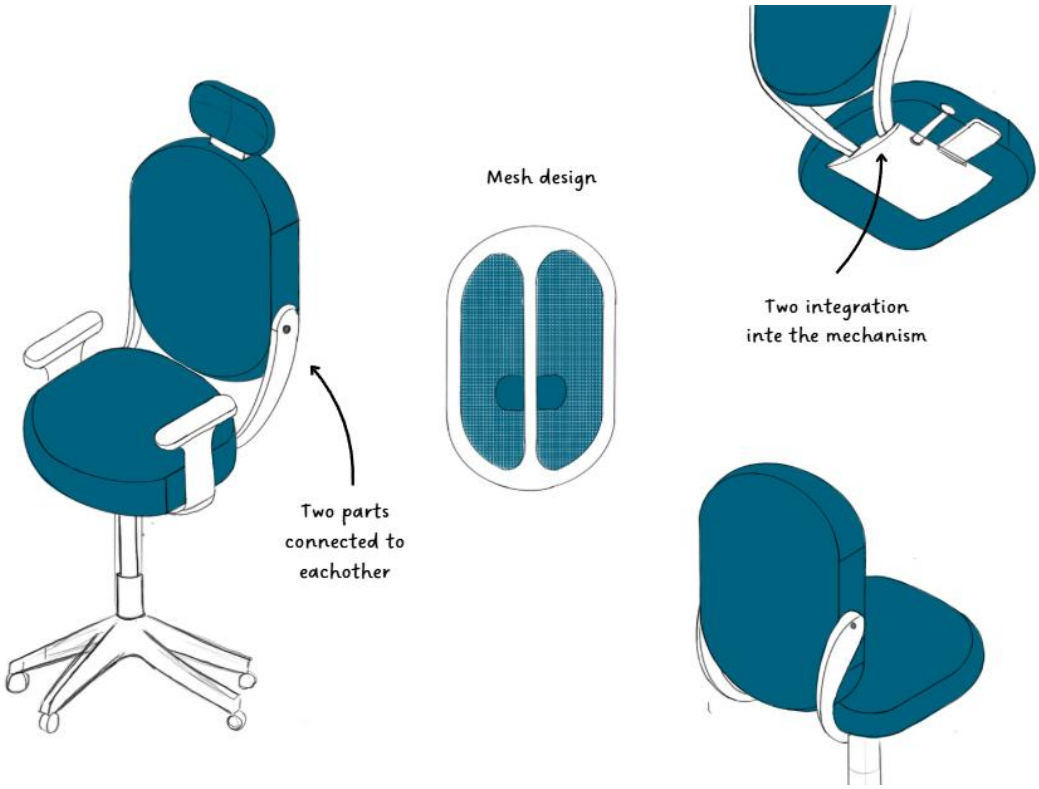



***Figure 25:*** *Digital sketches of Concept 3.*

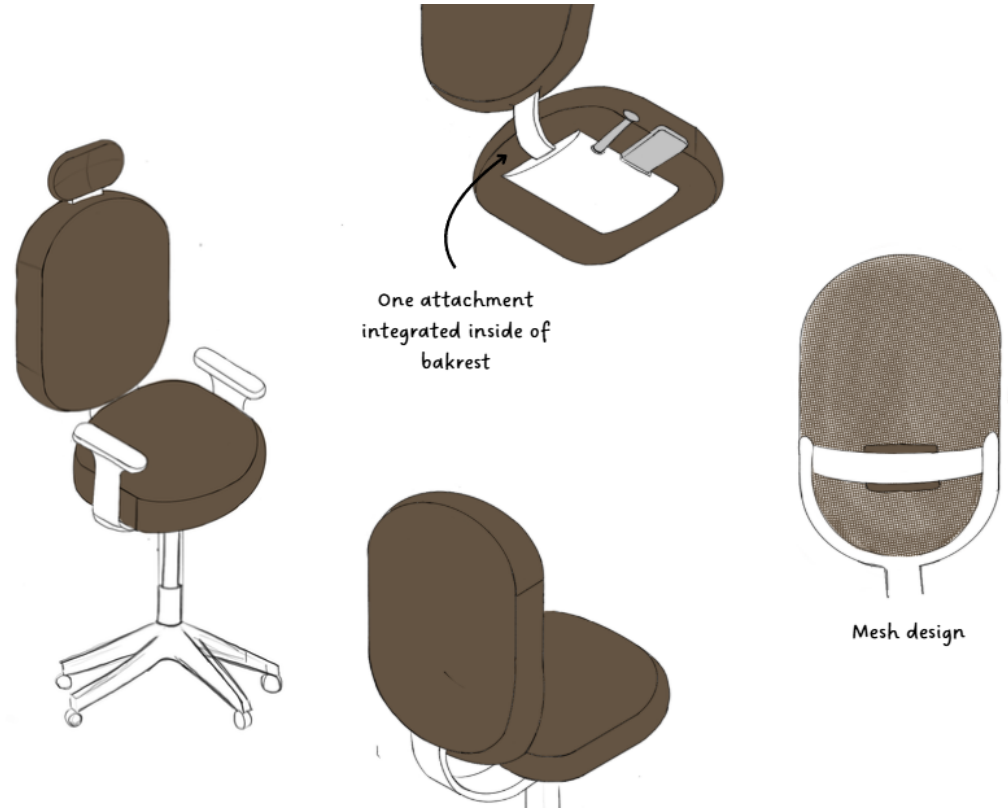



***Figure 26:*** *Digital sketches of Concept 4.*

After concept one had been selected, a second brainstorming phase was carried out to further develop the back view of the backrest attachment. Six sketches were created to explore different shape directions and how the attachment could visually connect to the backrest. The first shape was selected for further development, as its rounded form created a natural connection with the overall shape of the backrest.

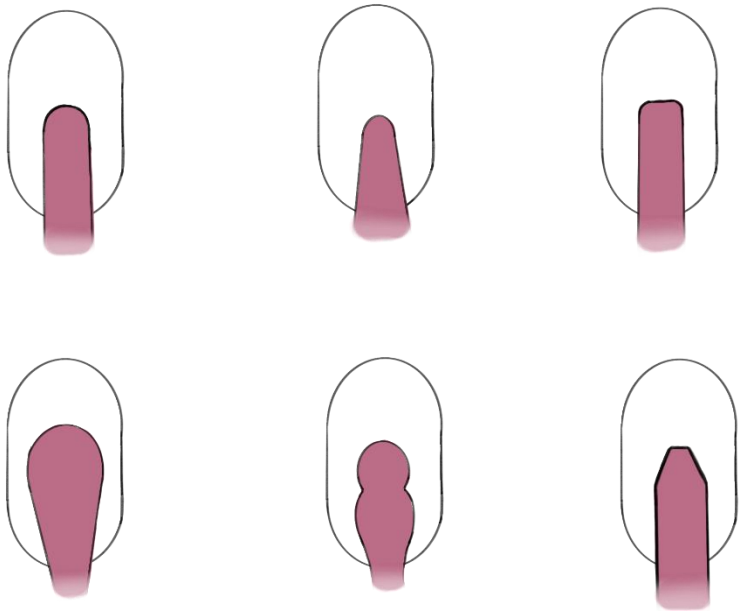


***Figure 27:*** *Further exploration of backrest attachment.*

Then the brainstorming became more focused on functional and technical details. CAD was used as a tool to place and evaluate the most important elements more precisely, especially the position of the adjustment controls and their relationship to the chair mechanism. This made it easier to understand how the buttons could be integrated under the seat in a practical and intuitive way. Three different control position ideas were developed. In the first idea, the seat height, backrest angle and rocking function were placed on the right side, while the seat depth adjustment was positioned at the front underside of the seat. In the second idea, the functions were divided into categories, with seat-related adjustments placed on the right side and backrest angle and rocking function placed on the left side. In the third idea, all adjustment functions were placed on the right side of the chair. The third idea was selected for further development. This direction was considered the most intuitive for the user and would suit the design language the best, since all functions are collected in one area instead of being spread across different sides of the chair. By placing the controls together, the user does not need to search in multiple places to find the adjustment functions. This supported the project aim of creating a chair that is not only functional, but also simple and easy to understand in use.

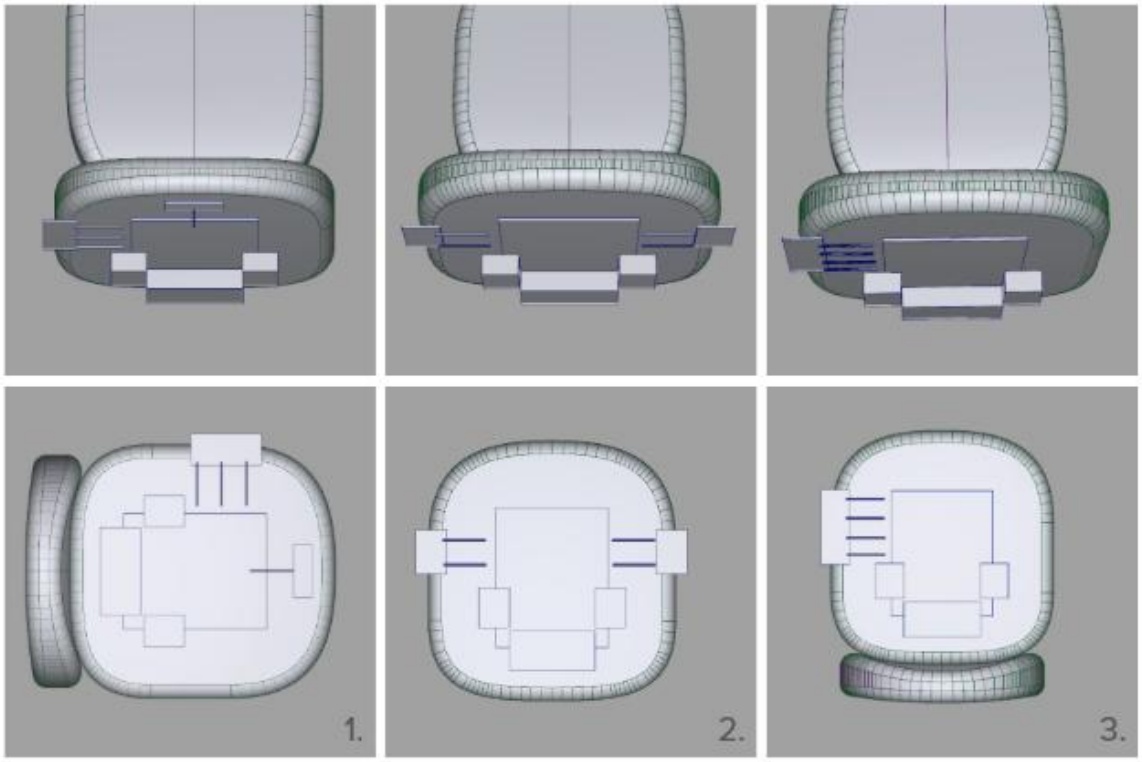



***Figure 28:*** *CAD exploration of different control positions for the backrest attachment.*

After the control positions had been explored, the development continued with the armrests. Different armrest designs were sketched by hand on an iPad using Procreate. The aim was to explore how the armrests could support the overall design language of the chair while still appearing simple, soft and integrated. Several variations were developed and compared in relation to the selected concept direction. Concept 6 was chosen for further development because it was considered to fit the design language best and supported the clean and modern expression of the chair.

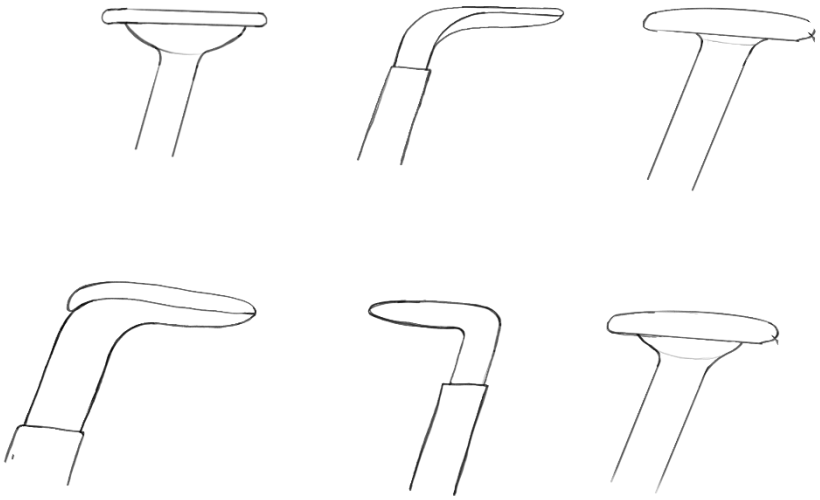


***Figure 29:*** *Sketches of six armrest design concepts, arranged from concept 1 in the upper left corner to concept 6 in the lower right corner.*

As a final part of the detailed concept development, a cover was designed using CAD modelling to hide the mechanism underneath the seat. Several cover concepts were created with slightly different forms and levels of coverage. During this exploration, the focus was to cover the mechanism without making the underside of the chair appear too bulky or heavy. Through several CAD iterations, a form was developed that balanced function and visual simplicity. Concept 5 was therefore selected as the final cover direction, since it covered the mechanism while still maintaining a clean and lightweight appearance.

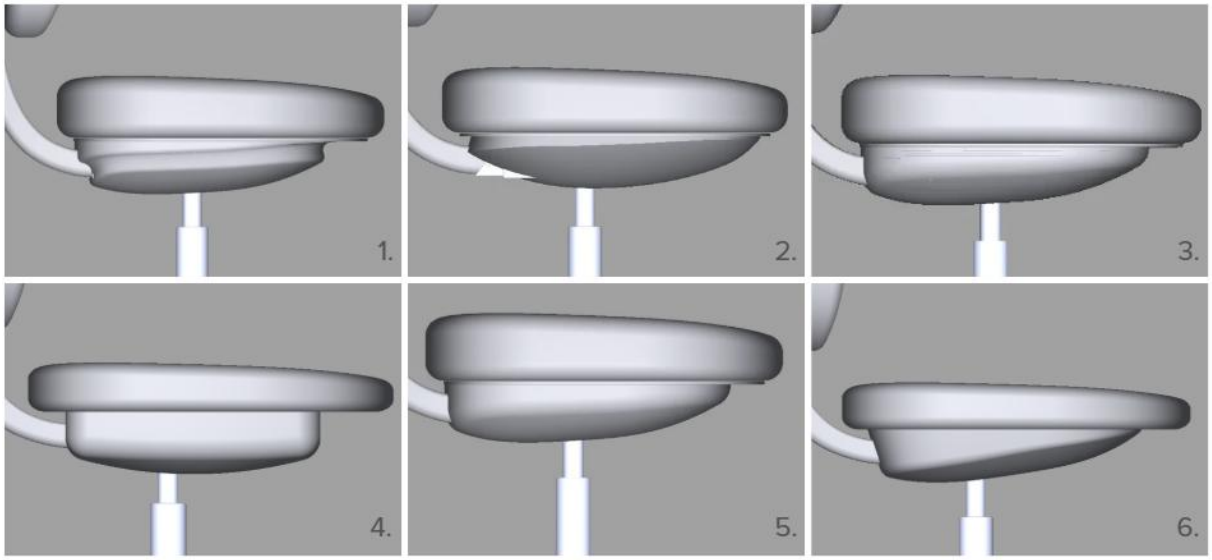



***Figure 30:*** *CAD exploration of six cover concepts for the underside of the seat, numbered from 1 to 3 in the top row and from 4 to 6 in the bottom row.*

## 4.4. Deliver

### 4.4.1. 3D physical model

Three physical models were created to support the presentation and evaluation of the final concept. The digital models were first developed using SolidWorks and Autodesk Alias. SolidWorks was used to create the CAD model of the platform and mechanism area, while Autodesk Alias was used to develop the overall chair design and form. These digital models were then prepared for 3D printing and printed using a Prusa MK4S printer.

The parts were printed in different colours to make the separate components easier to understand and visualize. Although the colours did not match the final colour palette exactly, they were close enough to communicate the intended material and design direction. This helped make the relationship between the platform, mechanism, seat and backrest clearer when presenting the concept.

One model was printed to show the mechanism, and the core platform functions in more detail. This model was printed with a scale factor of 50 and was used to visualize how the main ergonomic functions could be gathered in the shared platform. In addition, two smaller chair models were printed with a scale factor of 30. These models were used to show the overall proportions of the chair and how different chair variations could be created from the same platform.

The two smaller models represented two possible design variations: one chair with a mesh backrest and one chair with a foam-based backrest. By creating both versions, the physical models helped demonstrate how the shared platform could support different visual expressions and product variations, while keeping the same core structure. Overall, the 3D printed models made it easier to communicate the platform idea in a physical format and to show how the final concept could work as both a shared technical base and a design foundation for a chair collection.



***Figure 31:*** *3D printed prototypes of office chair design concept.*



***Figure 32:*** *3D printed prototypes of office chair design concept.*

# 5. Final concept result

## 5.1. Core platform

In the final concept, the platform is divided into a core platform and additional platform-related components. The core platform consists of the mechanism and the main connection points for the seat, backrest, armrests and control buttons. The backrest attachment is treated as a separate platform component, since it connects different backrest designs to the same core platform. Components such as armrests, headrests and different backrest types can then be added or varied depending on the chair version. This structure allows the platform to keep the main ergonomic and technical functions consistent, while still supporting design variation between different office chair models.

The mechanism was developed as the core platform of the final concept. It is placed underneath the seat and works as the main connection point for the chair's functional parts. The mechanism includes attachments for the backrest, armrests and control buttons, which makes it possible to gather several important functions in one shared base.  The base design is based Lanab groups already existing mechanism on Höganäs 280 but was modified and to fit the final platform concept. A new attachment was designed for the control buttons, where the shape was adapted to create space for the rods that connect the buttons to the different part inside the mechanism. The attachment is wider where the buttons are placed, providing enough space for the control rods and their connections. It then narrows toward the mechanism to fit the available surface on the mechanism and to avoid creating a bulky shape underneath the seat.

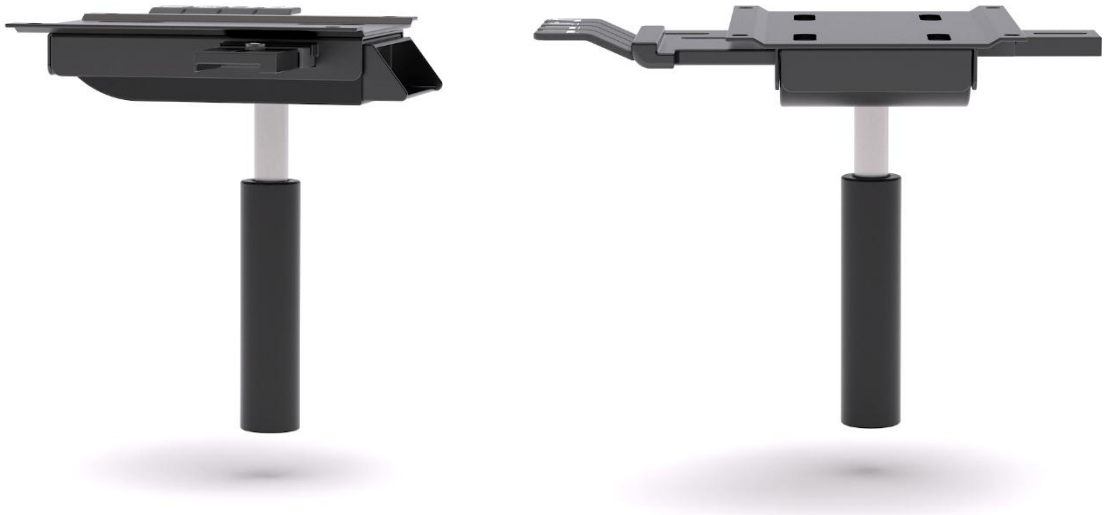


***Figure 33:*** *Final platform concept with core functions from side views.*

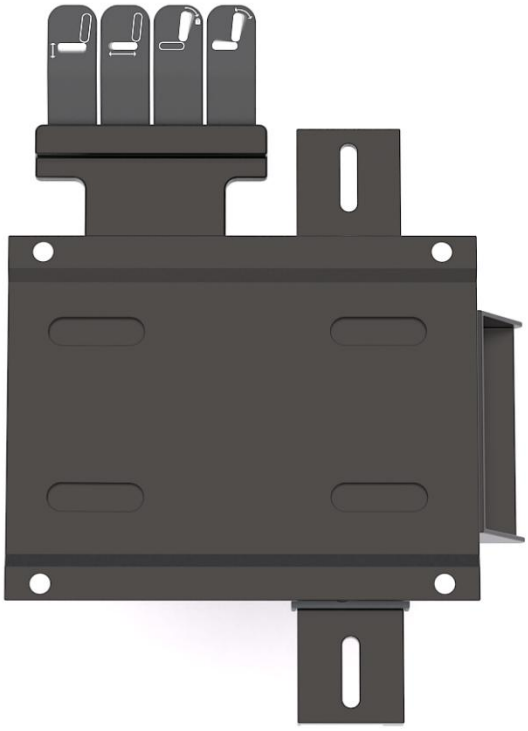


***Figure 34:*** *Final platform concept from top view.*

## 5.2. Backrest attachment

The backrest attachment is part of the platform, but not part of the core platform itself. It was designed as a separate platform component that connects the backrest to the mechanism. The design plays an important role in enabling variation and design variation. The final backrest attachment was designed as a shared component that can be mounted onto different backrest designs, such as the mesh chair and the more classic upholstered office chair shown in Figure 38.In this way, the backrest attachment allows the same core platform to be used across different chair variants, while the visible backrest design can change. The attachment connects into the mechanism and is mounted on the outside of the backrest. The conceptual solution also includes an adjustable backrest height function. This adjustment is controlled by a button placed on the backrest attachment. When the user pulls the button upward, the backrest is released into a free position, allowing it to be moved up or down. When the button is released, the backrest locks into the selected height. This allows the backrest to be adjusted to different users while keeping the same platform structure.

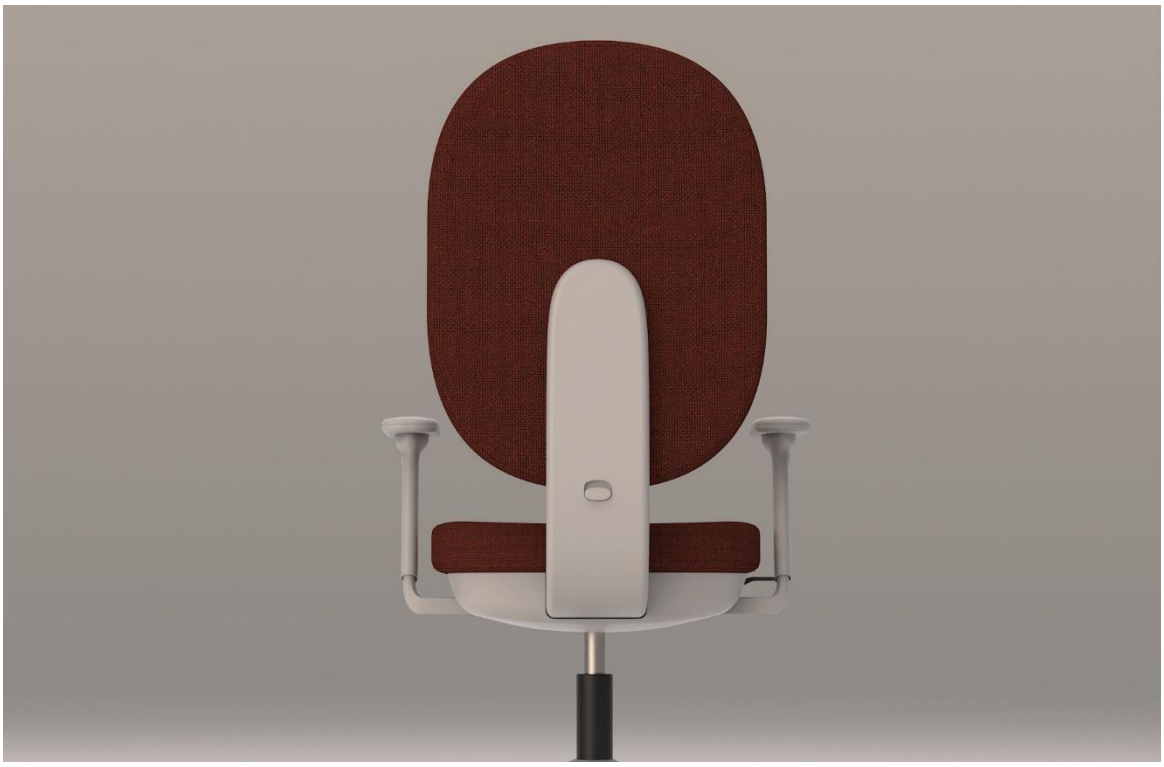


***Figure 35:*** *Backrest attachment.*

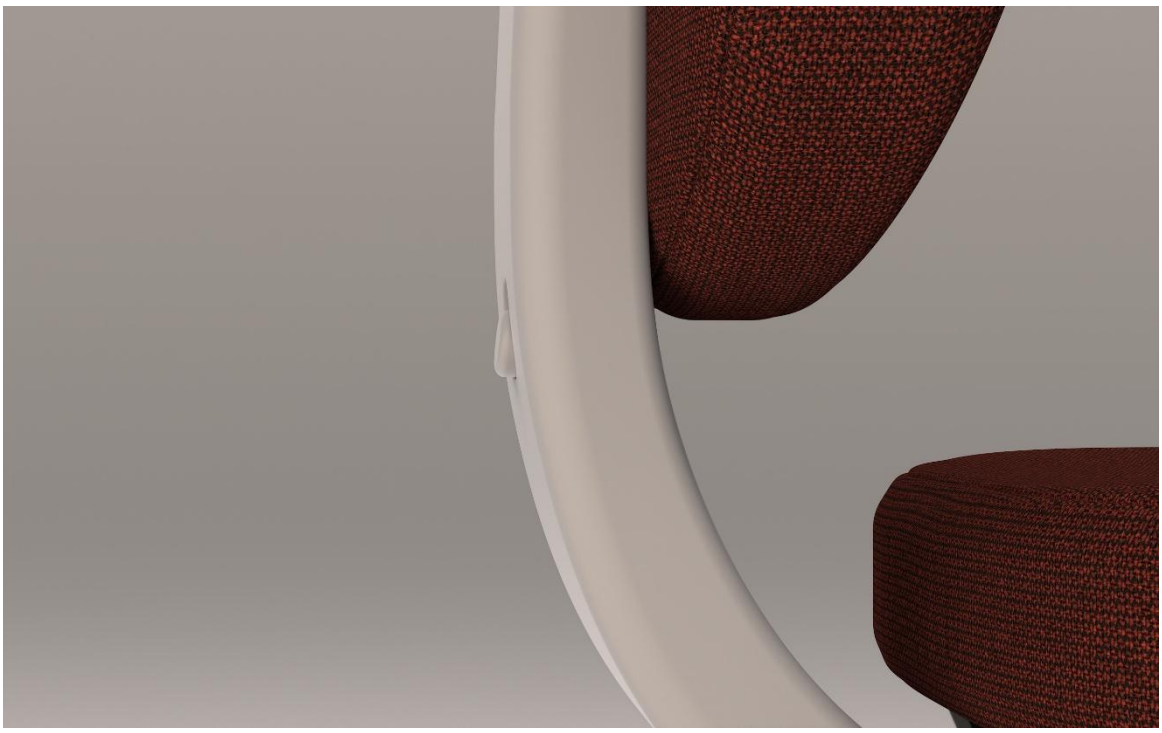


***Figure 36:*** *closeup on backrest height button*

## 5.3. Overall office chair design

The overall office chair design was guided by three keywords: *intuitive, minimal and soft.* These keywords are expressed in different ways in the final result of the chair design. The intuitive aspect is mainly expressed through the design and placement of the buttons. The result from the user observation showed the importance of intuitive controls, meaning that they should be easy to understand, find and use. Since the final concept includes several adjustable functions, the interaction with the controls became central for achieving an intuitive design.

All buttons were therefore designed to be easy to locate, operate and understand. The control buttons integrated into the core platform include graphic symbols that show what each function adjusts. This makes the ergonomic functions easier to understand for the user without needing additional instructions. The buttons are also placed on the right side of the chair and gathered close to each other, so the user can easily find and adjust the chair from one area. This reduces the risk of frustration caused by hidden or scattered controls. The button on the backrest attachment was designed slightly differently, since it controls the backrest height. Its shape is slightly extruded and angled upward to communicate that it is a movable part and that the user should pull it upward to release and adjust the backrest height.

To support the minimal design language, a cover was added underneath the seat to hide the mechanism and reduce the visual impression of technical components. This allows the platform to remain functional while giving the chair a cleaner and more integrated appearance. Placing the main control buttons on one side of the chair also contributes to the minimal expression, as it creates a more organised interaction area instead of having controls placed on several sides around the chair.

The soft expression is shown through the rounded shapes of the seat, backrest, backrest attachment and armrests. Large radii and smooth transitions were used to avoid sharp or heavy-looking forms and to create a more friendly and approachable appearance. The rounded edges also make the different parts feel more connected to each other, giving the chair a softer and more coherent overall expression



***Figure 37:*** *Final design concept from side and front view.*



***Figure 38:*** *final concept with mesh and classic backrest design.*



***Figure 39:*** *Render of final concept*

# 6. Discussion

This chapter discusses the results and reflects on how they relate to the overall aim of the study.

## 6.1. RQ1: Which ergonomic functions are essential when designing a shared product platform for office chairs?

In this project, an essential function is defined as a function that is necessary for the chair to fit the needs of office chair users. Based on the literature review, questionnaire, user observations, competitive benchmarking and function selection matrix, the essential functions identified for the shared office chair platform were seat height, seat depth, backrest height, lockable backrest tilt and rocking function. These functions can be seen as two groups: the first group focuses on body adaptation, while the second group focuses on postural variation and movement.

The first group includes seat height, seat depth and backrest height, which work together to make the chair suitable for different body types. Seat height mainly adjusts the vertical relationship between the user, the floor and the desk. Seat depths support different thigh lengths and influences how the user's legs and back relate to the seat and backrest. Backrest height allows the back support to be positioned according to the user's upper body height. The scoring result for this group showed that seat height and seat depth received the highest scores. However, backrest height did not score as highly as the other selected functions but was still included due to the strong support from the literature review and the findings from the user observations. The literature review showed that the position of the backrest is important for supporting the user's back correctly: if the backrest is too low, it may not support the intended area, while a backrest that is too high may feel misplaced or restrict the user. This becomes especially relevant when the same chair is expected to fit users with different body lengths.

The observations also supported this decision. Even if backrest height did not receive the strongest score, several users appeared to look for some form of adjustment or interacted with the chair as if they expected more adaptability. Many users also expressed discomfort due to the backrest position, either because it felt too high or too low, but they did not know that a backrest height adjustment existed. This suggests that the need for backrest height adjustment is related to body adaptation, even when it does not appear as a top-ranked function. For this reason, backrest height was understood as a platform-supported function, since it contributes to the ergonomic base of the platform but does not need to be integrated directly into the main mechanism underneath the seat. Instead, it can be solved through the backrest attachment.

Together, these three functions support the physical ergonomic perspective, since they help the same chair platform fit users with different body dimensions. This connects to the theoretical framework, where physical ergonomics is described as the relationship between the human body and the physical conditions of work, including posture, body dimensions and furniture design. The functions are also important in relation to the trend analysis and the interviews with purchasers, which showed that office environments are shifting towards more shared and flexible ways of working. Theactivity-based offices, shared desks and hybrid work environments, the same chair may be used by several people instead of being adapted to one fixed user. In this way, the

platform also relates to organizational ergonomics, since it supports a workplace structure where furniture needs to work for several users, routines and office settings. In a product platform context, this means that the shared platform needs to provide a flexible ergonomic base. Without these adjustments, the platform could become less suitable for the broad range of users found in today's office environments.

The second group of essential functions supports dynamic sitting, which also relates to physical ergonomics by allowing the body to move and vary posture during prolonged sitting. In the function selection matrix, lockable backrest tilt received a score of 4.40, while the rocking function received a score of 4.10. Although both functions received high score, both functions were included because they support postural variation in different ways. Lockable backrest tilt provides stable support by allowing the user to adjust and lock the backrest in a preferred position, which can be useful during focused work. The rocking function, on the other hand, supports small continuous movements and allows the user to shift between upright and reclined positions during the workday. Together, these functions support both stability and movement, making the platform more suitable for modern office work where users move between focused tasks, meetings and more informal work situations.

From a platform perspective, the selected functions can be understood on different levels. Seat height, seat depth, lockable backrest tilt and rocking function were considered core platform functions, since they are directly connected to the mechanism underneath the seat. Backrest height was also considered essential, but was understood as a platform-supported function, since it contributes to the shared ergonomic base but is solved through the backrest attachment rather than the main mechanism. Other functions, such as armrest variations, headrest adjustment, lumbar adjustment and more advanced comfort features, can still add ergonomic value, but were understood as addable functions/components, since they can be included depending on user needs, purchaser requirements or chair variant. This distinction also connects to organisational ergonomics, since the platform can keep the main ergonomic functions consistent while still allowing different chair versions to be adapted to different workplace needs, users and purchasing requirements.

However, the function selection should also be understood as a concept-level evaluation rather than an absolute ranking of ergonomic importance. The scoring matrix combined different types of data, including literature support, questionnaire results, observed use and competitor presence. These sources contributed with different perspectives, but they do not have the same meaning. For example, competitor presence shows what is common on the market, but it does not necessarily prove that a function is ergonomically essential. Similarly, observed use may be affected by whether users were able to find and understand a function during the observation. This means that some functions, such as lumbar support and armrest adjustment, may still have ergonomic value even if they were not included in the core platform. In this project, they were therefore not treated as unimportant, but as addable functions that can support variation depending on user needs, purchaser requirements and chair version.

## 6.2. How can an office chair be designed based on a shared product platform to support product variation?

The second research question focuses on how an office chair can be designed around a shared product platform while still allowing variation in form, material, components and expression. In this project, the platform was not only understood as a technical structure underneath the seat, but also as a design foundation that could connect several chair variants through the same ergonomic logic, visual language and functional layout. Therefore, the design of the office chair needed to balance multiple aspects: a shared platform, user-friendly interaction, and a design expression suitable for today's flexible and shared office environments.

### 6.2.1. Design language and user interaction

The design language and user interaction of the final concept were mainly shaped by the trend analysis and the user observations. The trend analysis showed that modern office environments are moving towards softer, more flexible and more welcoming spaces, while the user observations showed that chair adjustments need to be easy to locate, understand and use. Therefore, the design direction was based on an intuitive, friendly and minimal expression. This allowed the chair to respond to today's office environments while also supporting a more user-friendly interaction with the ergonomic functions.

As shown in the final concept, the softness was expressed through rounded forms, soft transitions and a visually calmer platform area. These design choices allow the chair to fit into modern office interiors while keeping a coherent design identity across variants. The cover presented in the final concept, helped express the minimal expression. By covering the mechanism, the platform becomes part of the design expression rather than only a technical component. The intuitive aspect of the design was especially important in relation to the adjustment controls. The observations showed that participants first looked for the adjustment functions on the right side of the chair, which made this a logical placement for the main controls. As presented in the final concept, the controls were gathered and organised according to these user expectations. This is important because the platform does not only need to include the right ergonomic functions; it also needs to make them easy to locate, understand and adjust in practice.

At the same time, gathering several functions in one area creates a risk that the controls become difficult to separate from each other. To reduce this, the buttons were supported by graphic symbols that communicate which function each control is connected to. In this way, the control design supports both physical ergonomics and cognitive ergonomics: the functions support the body, while the control layout helps the user understand and activate those functions in practice.

From an ergonomic perspective, this design approach can support cognitive ergonomics. The final concept shows that the ergonomic value of the platform depends not only on which functions are included, but also on whether users can locate, understand and use them in practice. By gathering the controls in an expected area and supporting them with graphic symbols, the design reduces uncertainty, trial-and-error interaction and mental effort when adjusting the chair. In this way, the user interaction supports the platform by making the ergonomic functions more accessible and understandable.

### 6.2.2. Platform structure and product variation

The platform structure supports product variation by separating the chair functions and components into different platform levels: core platform functions, platform-supported function and addable functions. The core platform functions, identified in RQ1, include seat height, seat depth, lockable backrest tilt and rocking function. These functions are directly connected to the main mechanism underneath the seat and create the shared ergonomic base of the chair. Since these functions remain consistent across variants, the platform can support product variation without losing its essential ergonomic foundation.

Backrest height was understood as a platform-supported function. As discussed in RQ1, backrest height has ergonomic value because it helps the chair adapt to different upper body lengths and dimensions. However, from a platform design perspective, it does not need to be integrated directly into the main mechanism in the same way as seat height, seat depth or tilt functions. Instead, it can be solved through the backrest attachment. This means that backrest height still contributes to the physical ergonomic quality of the chair, while allowing the main mechanism to remain shared across different variants.

Around this shared ergonomic base, addable functions and components can be used to create product variation. These may include armrest variations, headrests, lumbar support, different backrest types, upholstery, materials and colour options. This allows different chair versions to be developed from the same platform without changing the core platform functions. For example, one variant could include more advanced lumbar support and a mesh backrest, while another could focus on a softer upholstered expression with fewer visible adjustments. In both cases, the same core platform can remain underneath.

This structure shows how the office chair can be designed as a flexible product system rather than as one fixed product. The core platform functions provide consistency, while the addable components allow variation in expression, comfort level and functional configuration. From a physical ergonomic perspective, this is important because the essential body-supporting and movement-supporting functions remain present across variants. From a cognitive ergonomic perspective, it is also important that the core functions and control logic remain consistent, since this can make different versions of the chair easier to understand and use. Even if the visible design or component level changes, the user can still interact with the main ergonomic functions in a similar way.

The platform structure also relates to organizational ergonomics, since modern office environments often include shared desks, activity-based workspaces and different types of users within the same workplace. In these contexts, a chair collection may need to respond to different workplace settings, purchaser requirements and user needs. A shared platform with addable components makes it possible to create several chair variants for different contexts, while keeping the same basic ergonomic structure. This supports the idea that product variation can be achieved around a common platform rather than by designing each chair as a separate product.

However, one limitation of the proposed platform design is the cover underneath the seat. The cover contributes to the minimal and clean expression of the chair, but it is also designed specifically for this concept and therefore needs to match the form of the seat. This means that if the same mechanism is used with another seat shape, the cover may need to be adapted. Since production cost was not evaluated in this thesis, this should mainly be understood as a design limitation rather than a confirmed production issue. A similar aspect can be seen in the

attachment between the seat, backrest and mechanism. In this study, the attachment was designed with a rounded visual expression to fit the overall design direction. However, if the connection points and dimensions towards the main mechanism remain the same, the visible design of the attachment could be adapted and redesigned for other chair variants. This shows both a limitation and an opportunity within the platform: some external parts may need to change between variants, but they can also be used to create different visual identities while the same technical platform remains underneath.

# 7. Conclusion

This chapter summarises personal reflection and presents suggestions for future development.

## 7.1. Future scope

The result of this project presents a conceptual proposal for an office chair product family based on a shared product platform. The concept explores how essential ergonomic functions can be integrated into a common chair architecture while still allowing variation in expression, materials and additional components. The final concept should therefore not be seen as a finished product ready for production, but as a design proposal that shows how a platform-oriented approach could be applied within Lanab Group AB's product context.

A central part of the concept is the chair mechanism, since it connects several important components such as the seat, backrest, gas lift, armrests and base. In this project, Lanab Group AB's existing mechanism and measurements were used as a foundation for the overall placement and proportions of the chair. However, the mechanism itself was not technically developed in detail. The focus was instead placed on how the mechanism could support the platform structure and how the user-facing controls could be integrated into the design in a more intuitive and visually coherent way. Further development is therefore needed before the concept could be moved towards production. One important area is the technical development of the mechanism. Since the mechanism acts as the central connection point in the platform, its construction, dimensions, movement, load capacity and component interfaces would need to be further investigated. The placement and function of the control buttons would also need to be developed in more detail to ensure that they are technically feasible, durable and easy to use.

Future work should also include physical prototyping and user testing. A prototype would make it possible to evaluate the proportions, comfort, adjustment functions and overall user experience in a more realistic context. User testing and validating would be especially important to understand whether the controls are intuitive and accessible, and whether the ergonomic functions support users in practice. Manufacturing methods, assembly logic, material choices and cost would also need to be investigated further if the concept were to be developed into a production-ready product.

## 7.2. Personal reflection

This project has been a challenging and valuable experience that has given me a deeper understanding of product development, ergonomics and product platform thinking. The project required several different aspects to be considered at the same time, including user needs, ergonomic functions, product variation, visual expression, technical limitations and the company's existing product context.

At the beginning of the project, I assumed that the office chair could be designed as one complete product, where the overall form could be developed through only a few design sessions. However, after several brainstorming and design development sessions, it became clear that each part of the chair needed to be explored separately. The seat, backrest, armrests, mechanism, base and controls all had their own functional requirements, proportions and design challenges. This made the design development more time-consuming than

expected, but it also resulted in a more considered and coherent final concept. Another challenge was balancing the many perspectives involved in an office chair. The chair does not only need to meet the needs of the user in terms of comfort, ergonomics and intuitive use, but also needs to be relevant from a purchasing and interior design perspective. Trying to balance these perspectives made the project complex, but also more realistic and valuable.

Due to the scope of the project and the available time frame, more focus was placed on research, pre-study and concept development. Within the time available, the project included research, analysis, ideation and development of the main parts of the chair. However, I would have liked the project to also include physical prototyping, testing and evaluation. These steps would have made it possible to further improve the comfort, usability and technical feasibility of the concept.

Lastly, I would like to thank my classmates who have been part of my journey and supported me throughout the project. I would also like to give a special thanks to my supervisor at the university and my supervisor at Lanab Group AB for their time, guidance and valuable advice during the process.

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# Appendix

## Appendix A- Complete scoring matrix

Due to the size of the complete scoring matrix, the full table is provided as a digital appendix through the link below. The file contains the complete evaluation process, including criteria, ratings, weighting and final scores used in the concept selection process.

Appendix_A__Benchmark_analysis_matrix.xlsx